Biology

Class 11 (English Medium)

Solved Exercises — Sindh Textbook Board

14 chapters — 329 questions

Chapter 1: Biological Molecules

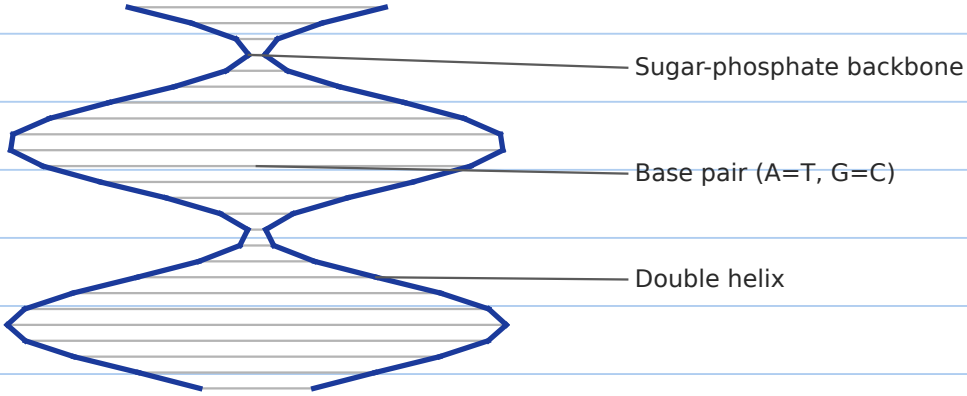


Fig: DNA Double Helix

1. Encircle the correct choice

(i) The slight negative charge at one end of one water molecule is attracted to the slight positive charge of another water molecule. What is this attraction called?

(a) Covalent bond

✓ (b) Hydrogen bond

(c) Ionic bond

(d) Hydrophilic bond

Reason: The weak attraction between the partial positive hydrogen of one water molecule and the partial negative oxygen of another is a hydrogen bond.

(ii) Tendency of water to coalesce oil drop into large droplet called

(a) Hydrophilic force

✓ (b) Hydrophobic exclusion

(c) Hydrophilic exclusion

(d) Hydrogen bonding

Reason: Non-polar oil molecules are excluded by water and come together, which is called hydrophobic exclusion.

(iii) The covalent bond between two monosaccharides is called

(a) Peptide bond

(b) Ester bond

(c) Phosphodiester bond

✓ (d) Glycosidic bond

Reason: Two monosaccharides are joined by a glycosidic bond formed by condensation.

(iv) Most abundant carbohydrate found in nature?

(a) Glucose

(b) Maltose

✓ (c) Cellulose

(d) Glycogen

Reason: Cellulose is the main structural carbohydrate of plant cell walls and is the most abundant carbohydrate in nature.

(v) Most important organic compound of the cell which carry out virtually all of the cell's activities.

✓ (a) Protein

(b) Carbohydrates

(c) Nucleic acid

(d) Lipids

Reason: Proteins work as enzymes, hormones, transporters and structural parts and carry out almost all cell activities.

(vi) All amino acids have same formula except

(a) Alpha carbon

(b) Hydroxyl group

✓ (c) Radical group

(d) Amino group

Reason: Every amino acid has the same amino group, carboxyl group and alpha carbon; only the radical (R) group differs.

(vii) A trihydroxy alcohol, made of three carbon atoms called

(a) Glucose

✓ (b) Glycerol

(c) Maltose

(d) Ribose

Reason: Glycerol is a three-carbon alcohol with three hydroxyl groups and forms the backbone of acylglycerols.

(viii) Large and important class of lipid made up of isoprenoid unit called

(a) Phospholipid

✓ (b) Terpenoids

(c) Waxes

(d) Acylglycerol

Reason: Terpenoids are a large class of lipids made of isoprenoid units (C5H8).

(ix) How many molecules of water are needed to completely hydrolyze a polymer that is 21 monomers long?

(a) 10

✓ (b) 20

(c) 21

(d) 2

Reason: A polymer of 21 monomers has 20 bonds between them, so

20 water molecules are needed to break all the bonds.

(x) Which of the following is true of both starch and cellulose?

(a) They are both polymers of glucose

(b) They are geometric isomers of each other.

(c) They can both be digested by humans

(d) They are both used for energy storage in plants.

Reason: Both starch and cellulose are polymers of glucose; they differ in the type of glycosidic linkage.

2. Write short answers of the following questions:

Question: 1. Why water molecules are called amphoteric in nature?

Answer: Water is called amphoteric because it can behave both as an acid and as a base. Due to ionization, a water molecule can donate a hydrogen ion (H+) and act as an acid, or accept a hydrogen ion and act as a base. This property helps water maintain pH and take part in many biochemical reactions in the cell.

Question: 2. Why amino acids are named so?

Answer: Amino acids are named so because each molecule contains both an amino group (-NH2) and a carboxylic acid group (-COOH). These two groups are attached to the same central alpha carbon atom. The amino group gives basic character and the carboxyl group gives acidic character, so the name combines both groups.

Question: 3. How monosaccharides are classified?

Answer: Monosaccharides are classified on the basis of the number of carbon atoms and the type of carbonyl group. According to carbon number they may be trioses (3C), tetroses (4C), pentoses (5C) like ribose, and hexoses (6C) like glucose. According to the carbonyl group they are

aldoses, having an aldehyde group, or ketoses, having a ketone group.

Question: 4. Enlist bio-elements make 98 % of living system?

Answer: Six bio-elements make about 98% of the living system. They are carbon, hydrogen, oxygen, nitrogen, phosphorus and sulphur. These elements form carbohydrates, proteins, lipids and nucleic acids. The remaining elements like calcium, potassium, sodium, chlorine, magnesium and iron are present in small amounts but are also essential.

Question: 5. Why fats provide more energy than carbohydrates?

Answer: Fats provide more energy than carbohydrates because fat molecules contain a much higher proportion of carbon-hydrogen bonds and very little oxygen. On complete oxidation, one gram of fat gives about 9 kcal of energy, while one gram of carbohydrate gives about 4 kcal. Fats are also stored in a water-free form, so they store more energy in less space.

Question: 6. How many steps involve in nucleotide formation?

Answer: Nucleotide formation involves joining of three components in steps. First a nitrogenous base is attached to a pentose sugar (ribose or deoxyribose) to form a nucleoside. Then one or more phosphate groups are attached to the sugar to make a nucleotide. Thus base + sugar + phosphate are combined step by step to complete a nucleotide.

Question: 7. Distinguish between saturated acylglycerol and unsaturated acylglycerols.

Answer: Saturated acylglycerols have fatty acids with only single bonds between carbon atoms, so they are fully loaded with hydrogen. They are usually solid at room temperature, e.g. animal fats. Unsaturated acylglycerols have one or more double bonds in their fatty acids, which

cause bends in the chains. They are usually liquid at room temperature, e.g. plant oils.

3. Write detailed answers of the following questions:

Question: 1. Describe the properties and roles of starch, glycogen, cellulose and chitin.

Answer: Starch is the storage polysaccharide of plants, made of glucose units, and it stores energy in seeds and tubers which can be broken down when needed. Glycogen is the storage polysaccharide of animals, highly branched, stored in liver and muscles and quickly converted to glucose. Cellulose is a structural polymer of glucose with beta linkages; it forms plant cell walls and gives strength and support, and humans cannot digest it. Chitin is a structural polysaccharide containing nitrogen, found in fungal cell walls and the exoskeleton of arthropods, where it provides protection and support.

Question: 2. What is nucleic acid? Describe structure of a mononucleotide (ATP) and a dinucleotide (NAD).

Answer: Nucleic acids are polymers of nucleotides that store and transfer genetic information; DNA and RNA are the two types. ATP is a mononucleotide consisting of adenine base, ribose sugar and three phosphate groups; its high-energy phosphate bonds store and release energy for cell activities. NAD is a dinucleotide formed by joining two nucleotides, one containing adenine and the other nicotinamide, linked through their phosphate groups. NAD works as a coenzyme and carries hydrogen and electrons during respiration.

Question: 3. What is amino acid? Explain peptide linkage formation.

Answer: An amino acid is an organic compound containing an amino

group, a carboxyl group, a hydrogen atom and a variable R group attached to a central alpha carbon. About 20 different amino acids make all proteins. A peptide linkage is formed when the carboxyl group of one amino acid reacts with the amino group of another amino acid. A molecule of water is removed in this condensation reaction, and the resulting -CO-NH- bond is the peptide bond. Repeating this process joins many amino acids into a polypeptide chain.

Question: 4. Explain classification of protein and list structural and functional protein.

Answer: Proteins are classified on the basis of structure into fibrous and globular proteins, and on the basis of composition into simple, conjugated and derived proteins. They are also classified by function. Structural proteins build body parts, e.g. collagen in connective tissue, keratin in hair and nails, and elastin in ligaments. Functional proteins perform specific jobs, e.g. enzymes catalyse reactions, haemoglobin transports oxygen, antibodies give protection and hormones like insulin regulate body activities.

Question: 5. What are lipids? Explain acylglycerol and waxes

Answer: Lipids are a diverse group of organic compounds that are insoluble in water but soluble in organic solvents, and they include fats, oils, phospholipids and steroids. Acylglycerols are formed by combining glycerol with one to three fatty acids; triglycerides store energy, provide insulation and protect organs. Waxes are esters of long-chain fatty acids with long-chain alcohols. They form waterproof coatings on leaves, fruits, skin and feathers, reducing water loss and giving protection.

Question: 6. Explain terpenoids and its types.

Answer: Terpenoids are a large and important class of lipids made up of repeating isoprenoid units (C5H8). They are classified according to the number of these units, such as monoterpenes, sesquiterpenes, diterpenes, triterpenes and tetraterpenes. Carotenoids, which are tetraterpenes, act as photosynthetic pigments and give colour to carrots and leaves. Other terpenoids include rubber, steroids precursors, essential oils and vitamins A, E and K, showing their wide biological importance.

Question: 7. What are conjugated molecules? Explain types of conjugated molecules.

Answer: Conjugated molecules are formed when biomolecules of two different groups combine chemically and act as one unit. Glycolipids are conjugates of lipids and carbohydrates, found in brain white matter, myelin sheath and membranes. Glycoproteins are conjugates of carbohydrate and protein and work as receptors, transport proteins and blood group antigens. Lipoproteins are conjugates of lipids and proteins that transport lipids in blood plasma, and nucleoproteins are conjugates of nucleic acid and basic protein, forming chromatin, chromosomes and ribosomes.

Chapter 2: Enzymes

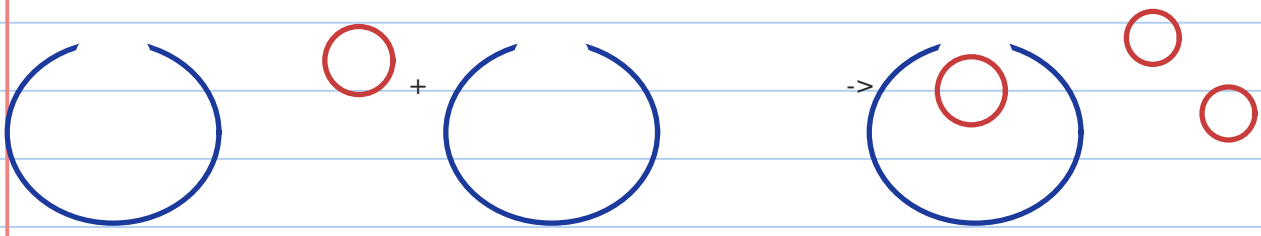


Fig: Enzyme Action (Lock and Key Model)

1. Encircle the correct choice

(i) Which of the following statements is (are) true about enzyme catalyzed reactions?

✓ (a) The reaction is faster than the same reaction in the absence of the enzyme.

(b) The free energy change of the reaction is opposite from the reaction in the absence of the enzyme.

(c) The reaction always goes in the direction toward chemical equilibrium.

(d) Both a and b.

Reason: Enzymes speed up a reaction by lowering activation energy, but they do not change the free energy change of the reaction.

(ii) The active site of an enzyme is the region that

(a) Binds allosteric regulators of the enzyme

✓ (b) Is involved in the catalytic reaction of the enzyme

(c) Binds the products of the catalytic reaction

(d) Is inhibited by the presence of a coenzyme or a cofactor.

Reason: The active site binds the substrate and takes part directly in the catalytic reaction.

(iii) According to the induced fit hypothesis of enzyme catalysis, which of the following is correct?

(a) The binding of the substrate depends on the shape of the active site.

(b) Some enzymes change their structure when activators bind to the enzyme.

(c) A competitive inhibitor can outcompete the substrate for the active site.

✓ (d) The binding of the substrate changes the shape of the enzyme's active site.

Reason: In induced fit, the substrate induces a change in the shape of the active site so that it fits the substrate more closely.

(iv) Which curve represents the behavior of an enzyme taken from a bacterium that lives in hot-springs. I. Curve 1 II. Curve 2 III. Curve 3

(a) I only

(b) II only

✓ (c) III only

Reason: Hot-spring bacteria have enzymes with a high optimum temperature, shown by Curve 3 which peaks at the highest temperature.

(v) Increasing the substrate concentration in an enzymatic reaction could overcome which of the following?

(a) Denaturation of the enzyme

(b) Allosteric inhibition

(c) Competitive inhibition

(d) Saturation of the enzyme activity

Reason: In competitive inhibition the inhibitor competes for the active site, so adding more substrate can overcome the inhibition.

(vi) Competitive inhibitors block the entry of substrate into the active site of an enzyme. On which of the following properties of an active site does this primarily depend?

✓ (a) The ability of an enzyme to form a template for holding and joining molecules

(b) The enzyme's ability to stretch reactants and move them toward a transition state

(c) The enzyme providing an appropriate microenvironment conducive to a reaction's occurrence

(d) The enzyme forming covalent bonds with the reactants

Reason: Competitive inhibition depends on the active site acting as a specific template whose shape the inhibitor mimics to block the substrate.

(vii) A series of enzymes catalyze the reaction X → Y → Z → A. Product A binds to the enzyme that converts X to Y at a position remote from its active site. This binding decreases the activity of the enzyme. What is substance X?

(a) A coenzyme

(b) An allosteric inhibitor

✓ (c) A substrate

(d) An intermediate

Reason: X is the starting material of the pathway, so it is the substrate of the first enzyme; product A is the allosteric (feedback) inhibitor.

✓

(viii) If an enzyme is added to a solution where its substrate and product are in equilibrium, what would occur?

(a) Additional product would be formed.

(b) Additional substrate would be formed.

(c) The free energy of the system would change.

✓ (d) Nothing; the reaction would stay at equilibrium.

Reason: An enzyme only speeds up reaching equilibrium; it cannot change the position of equilibrium, so nothing changes.

(ix) Which one of the following is the mechanism of action of enzymes?

(a) They act upon substrate molecules to release new substrate molecules.

(b) They actually increase the amount of energy of activation.

✓ (c) Enzymes dramatically decrease the amount of energy of activation.

(d) Enzymes break product molecules to release new product molecules.

Reason: Enzymes work by dramatically decreasing the activation energy needed for a reaction.

(x) How does a noncompetitive inhibitor decrease the rate of an enzyme reaction?

(a) By binding at the active site of the enzyme

✓ (b) By changing the shape of a reactant

(c) By changing the free energy change of the reaction

(d) By acting as a coenzyme for the reaction

Reason: A noncompetitive inhibitor binds away from the active site and changes the shape of the enzyme, so the reactant can no longer be acted upon effectively. (Note: it changes the enzyme's shape,

given in the book as changing the shape related to the reactant fit.)

2. Write short answers of the following questions:

Question: 1. Why enzymes are specific in nature?

Answer: Enzymes are specific because each enzyme has an active site of a definite shape and chemical nature. Only a substrate with a complementary shape can fit into that active site and form an enzyme-substrate complex. This is explained by the lock and key and induced fit models. Therefore one enzyme generally catalyses only one type of reaction.

Question: 2. Why enzyme activity is directly proportional to enzyme concentration?

Answer: Enzyme activity increases with enzyme concentration because more enzyme molecules provide more active sites for the substrate. When substrate is available in excess, every increase in enzyme molecules increases the number of enzyme-substrate complexes formed per unit time. As a result, more product is formed and the rate of reaction rises in direct proportion, until the substrate becomes limiting.

Question: 3. Why enzymes are called temperature sensitive?

Answer: Enzymes are proteins, so their activity depends strongly on temperature. At low temperature they work slowly, and activity increases up to an optimum temperature, about 37°C for human enzymes. Above the optimum, high temperature breaks the bonds holding the enzyme's shape and denatures it, so activity falls rapidly. This is why enzymes are called temperature sensitive.

Question: 4. How enzyme reduces the energy of activation?

Answer: An enzyme reduces activation energy by binding the substrate at

its active site and holding it in the correct position and stressed form. This brings reacting groups close together and weakens the bonds that are to be broken. The substrate therefore reaches the transition state easily with less energy. As a result the reaction proceeds faster at body temperature.

Question: 5. How enzymes activity effect on substrate concentration?

Answer: At low substrate concentration, increasing the substrate increases the rate because more active sites are occupied. As concentration rises, more enzyme-substrate complexes form and product formation speeds up. After a certain point all active sites become saturated, and the rate reaches a maximum and stays constant. Further increase in substrate then has no effect on enzyme activity.

Question: 6. What do you mean by prosthetic group?

Answer: A prosthetic group is a non-protein part that is permanently and tightly attached to an enzyme and is essential for its activity. It may be an organic molecule or a metal ion. Unlike a coenzyme, it does not separate from the enzyme during the reaction. Haem group in some enzymes is a common example of a prosthetic group.

Question: 7. Differentiate between the activator and inhibitors.

Answer: Activators are substances that increase the activity of an enzyme, often by helping the substrate bind or by removing an inhibitor. Inhibitors are substances that decrease or stop enzyme activity. Competitive inhibitors compete with the substrate for the active site, while non-competitive inhibitors bind at another site and change the enzyme's shape. Thus activators speed up reactions while inhibitors slow them down.

3. Write detailed answers of the following questions:

Question: 1. What is enzyme? explain characteristics of enzyme.

Answer: An enzyme is a biological catalyst, usually a protein, that speeds up a specific chemical reaction in living organisms without being used up itself. Enzymes are specific in action, work in small amounts and remain unchanged at the end of the reaction. They need an optimum temperature and pH for maximum activity. They lower the activation energy of reactions and their activity can be increased by activators or decreased by inhibitors. Most enzymes are globular proteins with an active site where the substrate binds.

Question: 2. Describe the effect if temperature on the rate of enzyme action.

Answer: Temperature strongly affects the rate of enzyme action. At low temperature enzyme molecules have little kinetic energy, so fewer collisions occur and the rate is slow. As temperature rises, collisions increase and the rate roughly doubles with every 10°C rise until the optimum temperature is reached. For most human enzymes the optimum is about 37°C. Beyond the optimum, heat denatures the enzyme by destroying its three-dimensional shape, and the rate falls sharply to zero.

Question: 3. Describe mechanism of enzyme action.

Answer: In enzyme action, the substrate first binds to the active site of the enzyme to form an enzyme-substrate complex. The lock and key model says the active site has a fixed shape complementary to the substrate, while the induced fit model says the active site changes shape slightly on binding. Binding stresses the substrate bonds and lowers activation energy, converting the substrate into products. An enzyme-product complex is formed, the products are released, and the free enzyme is ready to be used again.

Question: 4. Describe the classification of enzymes.

Answer: Enzymes are classified on the basis of the reaction they catalyse into oxido-reductases, transferases, hydrolases, lyases, isomerases and ligases. Oxido-reductases carry out oxidation-reduction, transferases transfer a group, and hydrolases break bonds by adding water. Lyases break bonds without hydrolysis, isomerases rearrange atoms within a molecule, and ligases join molecules in condensation reactions. Enzymes are also classified by substrate, e.g. proteases act on proteins, lipases on fats, glycosidases on carbohydrates and nucleases on nucleic acids.

Question: 5. Describe enzymatic inhibition, its types and its significance.

Answer: Enzyme inhibition is the decrease or stoppage of enzyme activity by a substance called an inhibitor. In competitive inhibition the inhibitor resembles the substrate and competes for the active site, and it can be overcome by increasing substrate concentration. In non-competitive inhibition the inhibitor binds at an allosteric site, changes the enzyme shape and cannot be overcome by substrate. Feedback inhibition occurs when the end product inhibits the first enzyme of its pathway. Inhibition is significant because it regulates metabolism, is used in drug discovery and helps identify disease conditions.

Question: 6. Describe structure of enzyme.

Answer: Most enzymes are globular proteins made of one or more polypeptide chains folded into a specific three-dimensional shape. A small region called the active site is where the substrate binds; it has a binding site and a catalytic site. Some enzymes are simple proteins, while conjugated enzymes consist of a protein part called apoenzyme and a non-protein cofactor. The cofactor may be a metal ion, a coenzyme or a tightly bound prosthetic group, and the complete active form is called

holoenzyme.

Question: 7. Explain the effect of substrate and concentration on the rate of enzyme action.

Answer: Substrate concentration increases the rate of enzyme action up to a limit. At first, more substrate means more collisions and more enzyme-substrate complexes, so the rate rises steeply. Gradually the active sites become occupied and the rate rises more slowly. When all active sites are saturated, the enzyme works at maximum rate (Vmax) and extra substrate has no effect. Enzyme concentration also increases the rate directly, provided substrate is in excess, because more active sites become available for the reaction.

Chapter 3: Cell Structure and Function

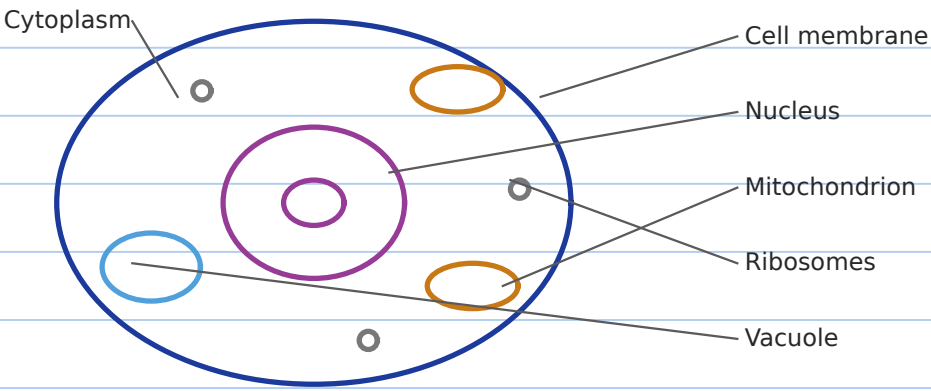


Fig: An Animal Cell

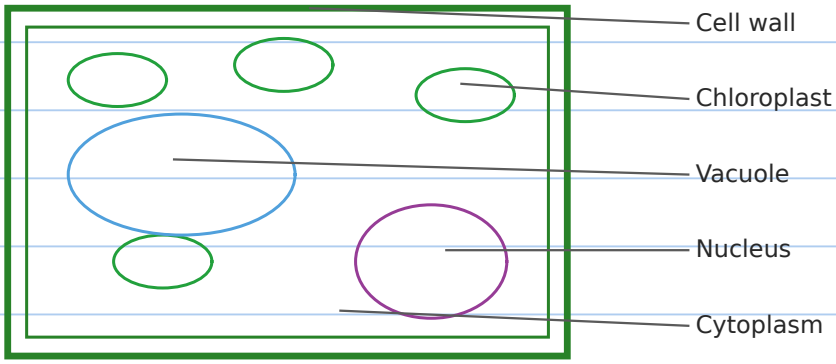


Fig: A Plant Cell

1. Encircle the correct choice

(i) A primary objective of cell fractionation is to

(a) View the structure of cell membranes.

(b) Identify the enzymes outside the organelles.

(c) Determine the size of various organelles.

(d) Separate the major organelles so that their particular functions can be determined. ✓

Reason: Cell fractionation separates the major organelles so that the particular function of each organelle can be studied.

(ii) The volume enclosed by the plasma membrane of plant cells is often much larger than the corresponding volume in animal cells. The most reasonable explanation for this observation is that

(a) Plant cells are capable of having a much higher surface to volume ration than animal cells.

(b) Plant cells have a much more highly convoluted plasma membrane than animal cells.

✓ (c) Plant cells contain a large vacuole that reduces the volume of the cytoplasm.

(d) Animal cells are more spherical, while plant cells are elongated.

Reason: Plant cells contain a large central vacuole that occupies most of the volume enclosed by the plasma membrane, making plant cells much larger in volume.

(iii) Large numbers of ribosomes are present in cells that specialize in producing which of the following molecules?

(a) Lipids

(b) Starch

✓ (c) Proteins

(d) Steroids

Reason: Ribosomes are the sites of protein synthesis, so protein-producing cells are rich in ribosomes.

(iv) In animal cells, hydrolytic enzymes are packed to prevent general destruction of cellular components. Which of the following organelles functions in this compartmentalization?

(a) Chloroplast

✓ (b) Lysosome

(c) Peroxisome

(d) Glyoxysome

Reason: Lysosomes pack hydrolytic enzymes in a membrane-bound sac so they digest only targeted material and do not destroy the whole cell.

(v) Tay-Sachs disease is a human genetic abnormality in cells accumulating and becoming clogged with very large and complex lipids. Which cellular organelle must be involved in this condition?

(a) Endoplasmic reticulum

(b) Golgi complex

✓ (c) Lysosome

(d) Mitochondria

Reason: In Tay-Sachs disease a lysosomal enzyme is missing, so complex lipids cannot be broken down and accumulate in cells.

(vi) Which is one of the main energy transformers of cells?

(a) Endoplasmic reticulum

(b) Golgi complex

(c) Lysosome

✓ (d) Mitochondria

Reason: Mitochondria transform energy from food into ATP, the usable energy form of the cell.

(vii) Organelles other than the nucleus that contain DNA I. Ribosomes II. Chloroplast III. Mitochondria

(a) I only

(b) II only

✓ (c) II and III

(d) I and II

Reason: Besides the nucleus, only chloroplasts and mitochondria contain their own DNA; ribosomes do not.

(viii) Which structure is common to plant and animal cells?

(a) Chloroplast

(b) Cell wall

(c) Central vacuole

✓ (d) Mitochondria

Reason: Mitochondria are present in both plant and animal cells, while chloroplast, cell wall and large central vacuole are plant features.

(ix) Cell organelle mainly concerned with the detoxification of alcohol.

(a) Chloroplast

✓ (b) Peroxisome

(c) Central vacuole

(d) Mitochondria

Reason: Peroxisomes are mainly concerned with the detoxification of alcohol and breakdown of hydrogen peroxide.

(x) Clarity of image is generally known as

(a) Magnification

(b) Contrast

✓ (c) Resolution

(d) Sedimentation

Reason: Resolution is the clarity of an image, i.e. the ability to distinguish two close points as separate.

2. Write short answers of the following questions:

Question: 1. Why lysosome is called suicidal sacs?

Answer: Lysosomes are called suicidal sacs because they contain powerful

hydrolytic enzymes capable of digesting the whole cell. When a cell is damaged or old, the lysosomal membrane may rupture and release these enzymes into the cytoplasm. The enzymes then digest the cell's own components, causing autolysis. This self-destruction is useful for removing worn-out cells and organelles.

Question: 2. Why plasma membrane is differentially permeable in nature?

Answer: The plasma membrane is differentially (selectively) permeable because its phospholipid bilayer allows only certain substances to pass. Small non-polar molecules pass easily, while ions and large polar molecules need specific transport proteins. This selective passage is due to the hydrophobic core of the bilayer and the presence of channel and carrier proteins. It helps the cell maintain its internal environment.

Question: 3. Why plant cell wall is rigid?

Answer: The plant cell wall is rigid because it is mainly made of cellulose microfibrils embedded in a matrix of pectin and other polysaccharides. Cellulose chains are held together by hydrogen bonds, forming strong bundles that resist stretching. In many cells lignin is also deposited, which makes the wall even harder. This rigidity gives shape, strength and protection to the plant cell.

Question: 4. Why chloroplast is called energy converting cell organelle?

Answer: Chloroplast is called an energy converting organelle because it converts light energy into chemical energy. During photosynthesis, chlorophyll absorbs sunlight and uses it to make ATP and NADPH2, which then fix carbon dioxide into glucose. In this way solar energy is changed into stored chemical energy of food. No other organelle performs this conversion in plants.

Question: 5. How prokaryotic ribosome is different from eukaryotic ribosome.

Answer: Prokaryotic ribosomes are smaller, of 70S type, made of 50S and 30S subunits. Eukaryotic ribosomes are larger, of 80S type, made of 60S and 40S subunits. Prokaryotic ribosomes are free in the cytoplasm, while eukaryotic ribosomes may be free or attached to the endoplasmic reticulum. This difference is also important in medicine, because some antibiotics target only bacterial 70S ribosomes.

Question: 6. How mitochondria is similar to bacteria?

Answer: Mitochondria resemble bacteria in several ways. They have their own circular DNA and 70S ribosomes like bacteria, and they can make some of their own proteins. They are surrounded by a double membrane and divide by a process similar to binary fission. Their size is also similar to that of bacteria, which supports the idea that mitochondria evolved from ancient bacteria.

Question: 7. Why mitochondria is called power house of cell?

Answer: Mitochondria are called the power house of the cell because they produce most of the cell's ATP during aerobic respiration. Glucose and other fuels are oxidized in the Krebs cycle and electron transport chain located in mitochondria. The energy released is stored in ATP, which powers all cell activities. Without mitochondria the cell would not get enough usable energy.

Question: 8. Differentiate between peroxisome and glyoxysome.

Answer: Peroxisomes are microbodies found in both plant and animal cells and are mainly concerned with detoxification of alcohol and breakdown of hydrogen peroxide by the enzyme catalase. Glyoxysomes are specialized

peroxisomes found only in plant cells, especially in germinating seeds. They contain enzymes that convert stored fatty acids into sugars needed for seedling growth. Thus glyoxysomes have a special role in seed germination, while peroxisomes work generally in detoxification.

3. Write detailed answers of the following questions:

Question: 1. Describe structure and functions of rough and smooth endoplasmic reticulum.

Answer: The endoplasmic reticulum is a network of tube-like channels of lipoprotein extending from the plasma membrane to the nuclear membrane. Rough ER has ribosomes attached to its surface, giving it a rough appearance, and it takes part in protein synthesis, folding and transport of proteins. Smooth ER has no ribosomes on its surface and is involved in lipid and steroid synthesis, detoxification and storage of calcium ions. Both types transport materials within the cell and provide a surface for biochemical reactions.

Question: 2. Explain the chemical composition and functions of plasma membrane in regulating cell's interactions with environment.

Answer: The plasma membrane is mainly composed of a phospholipid bilayer with proteins embedded in it, along with cholesterol and small amounts of carbohydrates. This fluid mosaic structure makes the membrane flexible and selectively permeable. It regulates the entry and exit of ions, nutrients and waste products through diffusion, osmosis and active transport. Membrane receptors receive signals from the environment and from other cells, while glycoproteins help in cell recognition. In this way the plasma membrane controls the cell's interactions with its environment.

Question: 3. Explain the structure and functions of lysosomes.

Answer: Lysosomes are small spherical sacs bounded by a single membrane and filled with hydrolytic enzymes. They are formed from the Golgi complex, and a newly formed lysosome before it starts its function is called a primary lysosome. Lysosomes digest food particles, worn-out organelles and invading bacteria and viruses. They also carry out autophagy and, when they rupture, autolysis of the cell. Their enzymes work best at acidic pH inside the lysosome, which protects the rest of the cell.

Question: 4. Describe structure of mitochondria with suitable diagram.

Answer: Mitochondria are rod-shaped organelles bounded by a double membrane. The outer membrane is smooth, while the inner membrane is folded into cristae that increase its surface area. The space inside the inner membrane is the matrix, which contains enzymes, circular DNA and 70S ribosomes. The cristae carry the electron transport chain and ATP synthase for oxidative phosphorylation. Because of their own DNA and ribosomes, mitochondria have a semi-autonomous existence in the cell. (Diagram to be drawn as in the textbook: outer membrane, inner membrane/cristae and matrix labelled.)

Question: 5. Explain the structure and composition of cell wall.

Answer: The plant cell wall lies outside the plasma membrane and is mainly composed of cellulose, pectin and other polysaccharides, with lignin in woody cells. The primary wall is thin and flexible, formed first, while the secondary wall is thicker and more rigid, deposited later. Adjacent cells are joined by the middle lamella made of pectin. The wall gives shape, mechanical strength and protection to the cell, prevents it from bursting, and still allows water and solutes to pass through it.

Question: 6. Describe the structure and functions of Golgi complex.

Answer: The Golgi complex consists of flattened membrane-bound sacs called cisternae, stacked together, with small vesicles around them. It is especially prominent in glandular cells. Products of the endoplasmic reticulum reach the Golgi complex, where they are modified, sorted, stored and packed into vesicles. These vesicles then send the products to their destination inside or outside the cell. The Golgi complex also forms lysosomes and secretes substances such as hormones, enzymes and mucus.

Question: 7. Describe the types, structure, composition and functions of cytoskeleton.

Answer: The cytoskeleton is a network of protein fibres that gives the cell its three-dimensional shape. Its main types are microfilaments made of actin, intermediate filaments, and microtubules made of tubulin protein. Microfilaments help in cell movement and muscle contraction, intermediate filaments provide mechanical strength, and microtubules form cilia, flagella, centrioles and the spindle during cell division. The cytoskeleton also helps in intracellular transport of organelles and vesicles. It constantly reorganizes itself according to the needs of the cell.

Chapter 4: Bioenergetics

1. Encircle the correct choice

(i) Chlorophyll-a is almost identical to chlorophyll-b but slight structural difference between them is enough to give

(a) Similar energy during the light reaction

✓ (b) Different absorptive spectrum

(c) Different product during the Calvin cycle

(d) All of these

Reason: The small structural difference between chlorophyll-a and chlorophyll-b gives them a different absorptive (absorption) spectrum.

(ii) Chlorophyll is organized along with other molecules into photosystem, which has light gathering

(a) Reaction Centre

(b) Carotenoid compound

✓ (c) Antenna complex

(d) Cytochrome

Reason: In a photosystem, chlorophyll and other pigments form the light-gathering antenna complex that passes energy to the reaction centre.

(iii) Select the correct statement

(a) PSI and ATP synthase complexes are located in the appressed part of thylakoid.

(b) PSI and NADP reductase are located in the appressed part of thylakoid membrane

(c) Appressed part Contain NADP reductase and ATP synthase

Reason: PSI is located in the non-appressed (non-stacked) regions of the thylakoid membrane, along with ATP synthase and NADP reductase.

(iv) The linear flow of electrons from water to NADP+ coupled to ATP synthesis is

(a) Cyclic photophosphorylation

✓ (b) Non-Cyclic photophosphorylation

(c) Chemiosmotic phosphorylation

(d) Oxidative phosphorylation

Reason: The linear flow of electrons from water to NADP+, coupled with ATP synthesis, is non-cyclic photophosphorylation.

(v) The intermediate carbon fixing compound in the member of grass family to pass CO_2 to calvin cycle is

(a) Citric acid

✓ (b) Oxaloacetic acid

(c) Pyruvic acid

(d) Crassulacean acid

Reason: In C4 members of the grass family, oxaloacetic acid is the first/intermediate carbon-fixing compound that passes CO_2 to the Calvin cycle.

(vi) Oxidative decarboxylation of isocitrate form?

✓ (a) α-ketoglutarate

(b) Succinate

(c) Cis-aconitate

(d) Fumarate

Reason: Oxidative decarboxylation of isocitrate in the Krebs cycle

forms α-ketoglutarate, releasing CO_2.

(vii) How many ATP molecules are forms during substrate level phosphorylation in kreb's cycle when one glucose is consumed?

(a) One

✓ (b) Two

(c) Three

(d) Four

Reason: One glucose gives two turns of the Krebs cycle, and each turn forms one ATP by substrate-level phosphorylation, so two ATP are formed.

(viii) Enzyme involves during carboxylation

(a) Rubisco oxygenase

✓ (b) Rubisco carboxylase

(c) Rubisco dehydrogenase

(d) No need of enzyme during carboxylation

Reason: Carboxylation in the Calvin cycle is carried out by the enzyme Rubisco carboxylase, which fixes CO_2 onto RuBP.

(ix) The oxygen consumed during cellular respiration is involved directly in which process or events?

(a) Glycolysis

✓ (b) Accepting electrons at the end of the electron transport chain

(c) The citric acid cycle

(d) The oxidation of pyruvate to acetyl CoA

Reason: Oxygen is consumed directly as the final electron acceptor at the end of the electron transport chain, forming water.

(x) How many carbon atoms are fed into the citric acid cycle as a result of

the oxidation of one molecule of pyruvate?

(a) 2

(b) 4

(c) 6

(d) 8

Reason: Oxidation of one pyruvate gives one acetyl CoA with 2 carbon atoms, which are fed into the citric acid cycle.

2. Write short answers of the following questions:

Question: 1. Why antenna complex contains other pigments with chlorophyll?

Answer: The antenna complex contains other pigments along with chlorophyll so that a wider range of light wavelengths can be absorbed. Accessory pigments like carotenoids and chlorophyll-b absorb light of different colours which chlorophyll-a alone cannot absorb well. They pass the absorbed energy to the reaction centre. This makes photosynthesis more efficient, especially in varying light conditions.

Question: 2. Why photosynthesis is called redox process?

Answer: Photosynthesis is called a redox process because it involves both oxidation and reduction. Water is oxidized, losing electrons and releasing oxygen. Carbon dioxide is reduced, gaining electrons and hydrogen, and is converted into glucose. Since oxidation and reduction occur together in the same process, photosynthesis is a redox process.

Question: 3. How cyclic photophosphorylation helpful in photosynthesis?

Answer: In cyclic photophosphorylation, electrons from Photosystem I return to it through electron carriers, and only ATP is produced. This extra ATP is needed because the Calvin cycle requires more ATP than NADPH2.

Cyclic flow therefore balances the supply of ATP and NADPH2 for carbon fixation. It also protects the plant when NADP+ is not available to accept electrons.

Question: 4. Why ATP is called common energy currency of living system?

Answer: ATP is called the common energy currency of living systems because it stores and supplies energy for almost all cellular activities. Energy from respiration and photosynthesis is trapped in ATP and later released by its hydrolysis to ADP wherever work is done. All organisms, from bacteria to humans, use this same molecule for energy transfer. Just like money, it is earned, stored for a short time and spent on needs.

Question: 5. Why Calvin cycle is also called C3 cycle?

Answer: The Calvin cycle is also called the C3 cycle because the first stable product of carbon fixation in it is a three-carbon compound. Carbon dioxide is fixed by RuBP with the help of Rubisco to form phosphoglyceric acid (PGA), which contains three carbon atoms. Plants using only this cycle are called C3 plants. Most common plants, such as wheat and rice, are C3 plants.

Question: 6. Why CAM plant close stomata in day time?

Answer: CAM plants close their stomata during the day to conserve water in hot and dry conditions. They open stomata at night, when evaporation is low, and fix CO2 into organic acids stored in vacuoles. During the day these acids release CO2 internally for the Calvin cycle while stomata remain closed. This adaptation greatly reduces transpiration and helps succulent plants survive in deserts.

Question: 7. Why oxidation of pyruvate provide more energy than lactic acid fermentation?

Answer: Oxidation of pyruvate in aerobic respiration completely breaks it down to CO_2 and water through the Krebs cycle and electron transport chain. This releases a large number of ATP molecules from one glucose molecule. In lactic acid fermentation, glucose is only partly broken down and lactic acid still contains much unused energy. Therefore aerobic oxidation of pyruvate provides far more energy than fermentation.

3. Write detailed answers of the following questions:

Question: 1. Explain in detail light independent phase of photosynthesis.

Answer: The light independent phase, or Calvin cycle, takes place in the stroma of the chloroplast and does not directly need light. In the first step, CO_2 is fixed by RuBP (5C) with the enzyme Rubisco to form PGA (3C). In the second step, PGA is reduced to glyceraldehyde phosphate using ATP and $NADPH_2$ produced in the light reactions. Some of this product is used to make glucose and other carbohydrates, while the rest regenerates RuBP so the cycle can continue. This phase depends on the products of the light phase but itself is light independent.

Question: 2. What is cellular respiration? Explain types of respiration in detail.

Answer: Cellular respiration is the breakdown of glucose by redox reactions to synthesize ATP, releasing energy for the cell. Aerobic respiration occurs in the presence of oxygen and completely oxidizes glucose to CO_2 and water, producing a large amount of ATP through glycolysis, Krebs cycle and electron transport chain. Anaerobic respiration occurs without oxygen; glucose is only partially broken down. Its types include alcoholic fermentation, producing ethanol and CO_2 in yeast, and lactic acid fermentation, producing lactic acid in muscles and some bacteria, with only 2 ATP per glucose.

Question: 3. Explain event takes place in breaking of glucose in cytosol.

Answer: The breaking of glucose in the cytosol is glycolysis, an anaerobic process common to almost all cells. One glucose molecule (6C) is first phosphorylated using ATP and then split into two molecules of pyruvate (3C). During the process a net gain of 2 ATP by substrate-level phosphorylation and 2 NADH2 molecules is obtained. Glycolysis does not require oxygen and takes place entirely in the cytosol. The pyruvate formed then enters mitochondria for aerobic respiration or is fermented in anaerobic conditions.

Question: 4. Discuss cyclic and non-cyclic photophosphorylation during light reaction.

Answer: In non-cyclic photophosphorylation, both photosystems II and I work in series; electrons flow linearly from water to NADP+, oxygen is released, and both ATP and NADPH2 are produced. In cyclic photophosphorylation, only Photosystem I takes part; electrons are passed back to it through carriers, so no oxygen is evolved and no NADPH2 is made, only ATP is formed. Cyclic flow supplies the extra ATP needed by the Calvin cycle. Both processes use light energy to make ATP by chemiosmosis across the thylakoid membrane.

Question: 5. Describes tricarboxylic acid cycle in detail.

Answer: The tricarboxylic acid (Krebs) cycle takes place in the mitochondrial matrix. Acetyl CoA (2C) combines with oxaloacetate (4C) to form citrate (6C), the first tricarboxylic acid. Through a series of steps involving oxidative decarboxylation, citrate is converted back to oxaloacetate, releasing 2 CO2. Per turn, 3 NADH2, 1 FADH2 and 1 ATP (by substrate-level phosphorylation) are produced. Since one glucose gives two turns, the yield is doubled, and the reduced coenzymes then feed the

electron transport chain.

Question: 6. Explain alternative mechanism of CO2 fixation in plant

Answer: Some plants avoid photorespiration by alternative CO2 fixation mechanisms. In C4 plants like sugarcane and maize, CO2 is first fixed in mesophyll cells by PEP into oxaloacetate (4C), which is converted to malate and transported to bundle sheath cells, where CO2 is released for the Calvin cycle. In CAM plants like cacti and pineapple, stomata open at night and CO2 is stored as organic acids in vacuoles, then released in the day for fixation. Both mechanisms keep CO2 concentration high around Rubisco and conserve water or increase efficiency.

Question: 7. Explain chemiosmosis and oxidative phosphorylation.

Answer: Chemiosmosis is the process in which ATP is synthesized using energy of a proton gradient across a membrane. During the electron transport chain, electrons from NADH2 and FADH2 pass through carriers and pump H+ ions, creating a gradient. Protons flow back through ATP synthase, and this flow drives the joining of ADP and phosphate to form ATP. Oxidative phosphorylation is this ATP formation linked with the oxidation of reduced coenzymes, with oxygen as the final electron acceptor. It produces most of the ATP of aerobic respiration in mitochondria.

Chapter 5: Acellular Life

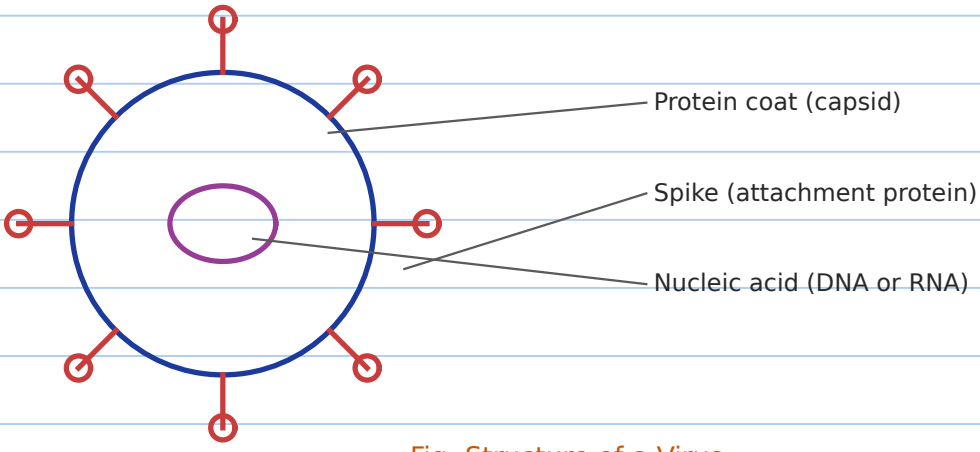


Fig: Structure of a Virus

1. Encircle the correct choice.

(i) Viruses evolve by using the

(a) Cellular organelles

(b) Cellular enzymes

(c) Cellular energy

✓ (d) All of them

Reason: Viruses have no machinery of their own, so they use the host cell's organelles, enzymes and energy for their reproduction and evolution.

(ii) The pathogenicity of a virus depends upon

(a) the immunity of the host body

✓ (b) the effective penetration of its genome

(c) the overall environment inside the body

(d) the overall environment outside the body

Reason: Pathogenicity depends on the effective penetration of the viral genome into the host cell, after which it takes over the cell machinery.

(iii) Bacteriophages escape from host cell by the activity of

✓ (a) Lysozyme

(b) Ribozyme

(c) Peroxisomes

(d) Glyoxisomes

Reason: Lysozyme made by the phage dissolves the bacterial cell wall, so new bacteriophages escape from the host cell.

(iv) The smaller proteins are cut down and forms a new virus structure by the process called

(a) Integration

(b) Transcription

(c) Budding

✓ (d) Assembly

Reason: In assembly, viral proteins are cut/processed and combined with viral genome to form new complete virus particles.

(v) Virus that severely damage motor neurons and causes paralysis called

(a) HIV

(b) Dengue

✓ (c) Polio

(d) Herpes

Reason: Poliovirus damages motor neurons of the spinal cord and causes paralysis, the disease poliomyelitis.

(vi) Proteins that cause pathogenicity in humans and animals called

✓ (a) Prions

(b) Viroids

(c) Antigen

(d) Antibodies

Reason: Prions are infectious protein particles that cause diseases in humans and animals, e.g. mad cow disease.

(vii) Smaller than viruses having single stranded RNA with some double stranded regions are called

(a) Prions

✓ (b) Viroids

(c) Minus strand viruses

(d) Double stranded DNA viruses

Reason: Viroids are subviral particles of single-stranded RNA with some double-stranded regions, smaller than viruses and without a protein coat.

(viii) Dengue fever, encephalitis and yellow fever are caused by which group of viruses?

✓ (a) Arbo-viruses

(b) Retro-viruses

(c) Rhabdo-viruses

(d) Rhino-viruses

Reason: These diseases are caused by arbo-viruses, which are transmitted by arthropods such as mosquitoes.

(ix) Aedes mosquito is the vector of

✓ (a) Dengue virus

(b) Ebola virus

(c) Hepatitis virus

(d) Measles virus

Reason: Aedes mosquito is the vector that transmits dengue virus

(and also chikungunya virus) through blood.

2. Write short answers of the following questions:

Question: 1. Discuss the living and nonliving status of virus

Answer: Viruses show both living and non-living characters. Outside a host they are inert particles that can be crystallized, have no metabolism and do not respire or grow, which are non-living characters. Inside a living host cell they reproduce, mutate and evolve using the host machinery, which are living characters. Because of this dual nature, viruses are placed at the boundary between living and non-living and are called acellular particles.

Question: 2. What do we mean by positive and negative sense virus?

Answer: A positive sense virus has single-stranded RNA that can act directly as messenger RNA and be translated into proteins by the host ribosomes. A negative sense virus has RNA complementary to mRNA, so it must first be transcribed into a positive strand by viral RNA polymerase before proteins can be made. This classification is based on the nature of the viral genome. Influenza virus is a negative sense virus, while polio virus is positive sense.

Question: 3. Name the groups of viruses from Baltimore classification

Answer: The Baltimore classification divides viruses into seven groups according to their genome and method of mRNA synthesis. Group I is double-stranded DNA viruses, Group II single-stranded DNA viruses, and Group III double-stranded RNA viruses. Group IV is positive sense single-stranded RNA viruses, Group V negative sense single-stranded RNA viruses, Group VI RNA retroviruses that use reverse transcriptase, and Group VII double-stranded DNA retroviruses. This system is widely used for virus classification.

Question: 4. How does bacteriophage virus infect bacteria?

Answer: A bacteriophage first attaches to specific receptors on the bacterial surface by its tail fibres. Its tail sheath contracts and the viral DNA is injected into the bacterium through the cell wall, while the protein coat stays outside. The viral DNA then takes over the bacterial machinery to make new phage parts. Finally the cell bursts and releases new phages, or the viral DNA may stay integrated in a lysogenic cycle.

Question: 5. Differentiate lytic and lysogenic life cycle of bacteriophage.

Answer: In the lytic cycle, the phage DNA immediately takes control of the bacterium, many new phages are produced, and the host cell is lysed (burst) to release them, so the host dies. In the lysogenic cycle, the phage DNA integrates into the bacterial chromosome as a prophage and is copied silently with it for many generations. The lysogenic cycle may later switch to the lytic cycle under certain conditions. Thus lytic cycle is destructive at once, while lysogenic cycle is initially harmless.

Question: 6. What is reverse transcription? How it performs by HIV in human?

Answer: Reverse transcription is the formation of DNA from an RNA template, the reverse of normal transcription. HIV carries the enzyme reverse transcriptase, which converts its single-stranded RNA into double-stranded DNA inside the host cell. This viral DNA enters the nucleus and integrates into the human chromosome as provirus. It then directs the production of new HIV particles, mainly in helper T lymphocytes.

Question: 7. List down any five animal and plant viruses with their vector transmission and symptom

Answer: Animal viruses: Dengue virus, transmitted by Aedes mosquito through blood, causes fever, rash and pain behind the eyes; Hepatitis C virus, transmitted by infected blood, causes fever, yellow skin and fatigue; Measles virus, spread by airborne droplets, causes fever, cough and body rash. Plant viruses include Tobacco mosaic virus, spread by contact, causing mosaic patterns on leaves, and Cotton leaf curl virus, transmitted by whitefly, causing curling of leaves and great loss of cotton yield. Each virus has a specific vector, mode of transmission and symptoms.

Question: 8. List down the sequences involved in the lytic and lysogenic life cycle of bacteriophage?

Answer: The lytic cycle sequence is: attachment to the host, penetration (injection of DNA), synthesis of phage components, assembly of new phages, and release by lysis of the host cell. The lysogenic cycle sequence is: attachment, penetration, integration of phage DNA as prophage into the bacterial chromosome, and replication of prophage with the host DNA during cell division. Under induction, the prophage may leave the chromosome and enter the lytic cycle. Both cycles begin the same way but differ after penetration.

Question: 9. How a virus survives without host discuss.

Answer: Outside a host, a virus survives as an inert particle called a virion, protected by its protein capsid and sometimes an envelope. It shows no metabolism and can remain inactive for long periods on surfaces, in air, water or soil, depending on temperature and humidity. Some viruses survive unfavourable conditions in dormant form inside host cells or seeds. When a suitable host is contacted, the virus becomes active again and starts its infection cycle.

Question: 10. What are the symptoms of AIDS?

Answer: AIDS is the final stage of HIV infection in which the immune system is badly damaged. Its symptoms include prolonged fever, night sweats, rapid weight loss, chronic diarrhoea and swollen lymph glands. The patient suffers repeated opportunistic infections such as pneumonia and tuberculosis, and skin lesions may appear. Because helper T cells are destroyed, even common infections become severe and life-threatening.

Question: 11. What is the prevention and control of AIDS?

Answer: AIDS can be prevented by avoiding contact with infected blood and body fluids. Sterilized or disposable syringes should be used, and blood must be screened before transfusion. Safe sexual practices and avoiding sharing of needles and razors are essential. An infected mother can reduce transmission to her baby by proper medical treatment during pregnancy and delivery. There is no complete cure, so prevention and awareness are the main methods of control.

Question: 12. Differentiate Prions and Viroids.

Answer: Prions are infectious particles made only of protein, with no nucleic acid, and they mainly cause diseases of the nervous system in humans and animals, e.g. mad cow disease. Viroids are infectious particles made only of small single-stranded RNA, with no protein coat, and they cause diseases only in plants. Prions were discovered by Stanley Prusiner and viroids by Theodor Diener. Both are subviral particles, simpler and smaller than viruses, but their chemical nature and hosts are different.

3. Write detailed answers of the following questions:

Question: 1. Explain the lytic life cycle of Bacteriophage with labelled diagram.

Answer: In the lytic cycle, the bacteriophage first attaches to the bacterial surface with its tail fibres. The tail contracts and viral DNA is injected into the cell, leaving the coat outside. The phage DNA takes over host machinery; host DNA is broken down and new phage DNA and proteins are synthesized. These parts are assembled into complete phages, and the enzyme lysozyme weakens the cell wall. Finally the bacterium bursts (lysis) and releases a large number of new phages to infect other cells. (Labelled diagram to be drawn as in the textbook: attachment, penetration, synthesis, assembly, release.)

Question: 2. Explain the life cycle of HIV with labelled diagram.

Answer: HIV attaches to CD4 receptors of helper T lymphocytes and enters the cell. Its RNA is converted into double-stranded DNA by reverse transcriptase, and this DNA integrates into the host chromosome as provirus. The provirus may remain latent or become active, producing viral RNA and proteins using host machinery. New HIV particles are assembled at the cell membrane and released by budding, acquiring an envelope. Gradually the helper T cells are destroyed, the immune system weakens, and AIDS develops. (Labelled diagram to be drawn as in the textbook.)

Question: 3. Explain the pathogenicity and economic losses caused by viruses to the humans.

Answer: Viruses cause diseases by penetrating host cells, multiplying inside them and damaging tissues, and their pathogenicity depends on host immunity and environmental conditions. Human diseases like influenza, hepatitis, dengue, polio, measles and AIDS cause millions of deaths and huge medical expenses. Epidemics reduce working capacity, close schools and industries, and burden hospitals, as seen during COVID-19 when production fell in most countries. Plant viruses such as

cotton leaf curl virus have caused losses of billions of rupees to Pakistan's cotton economy. Thus viruses damage both human health and the national economy.

Chapter 6: Prokaryotes

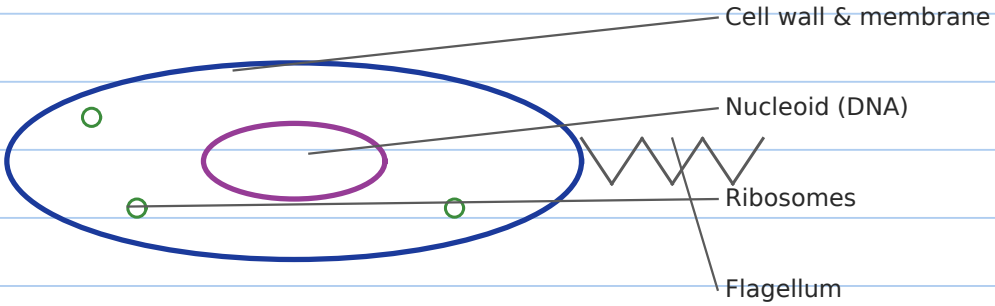


Fig: A Bacterial Cell

2. Write short answers of the following questions:

Question: 1. Explain the structure of bacteria in detail with labelled diagram.

Answer: A bacterium is a unicellular prokaryote surrounded by a cell membrane and a cell wall of peptidoglycan, sometimes with an outer capsule. Its cytoplasm contains a single circular DNA in the nucleoid region, plasmids, 70S ribosomes and storage granules, but no membrane-bound organelles. Many bacteria have flagella for movement, and pili or fimbriae for attachment and conjugation. Some form resistant endospores in unfavourable conditions. (Labelled diagram to be drawn as in the textbook: capsule, cell wall, membrane, nucleoid, plasmid, ribosome, flagellum.)

Question: 2. Explain the Autotrophic mode of nutrition in bacteria.

Answer: Autotrophic bacteria make their own food from simple inorganic substances. Photoautotrophic bacteria, such as cyanobacteria and green and purple sulphur bacteria, use sunlight as an energy source for photosynthesis. Chemoautotrophic bacteria obtain energy by oxidizing

inorganic substances like ammonia, nitrites, sulphur or iron, and use it to fix CO2. These bacteria are important in nutrient cycles, e.g. nitrifying bacteria in soil. They do not depend on other organisms for food.

Question: 3. Explain the photosynthesis mechanism in bacteria.

Answer: Bacterial photosynthesis uses light energy to make food, but most bacteria use bacteriochlorophyll instead of chlorophyll and do not release oxygen. In purple and green sulphur bacteria, hydrogen sulphide acts as the hydrogen donor, so sulphur is deposited instead of oxygen being evolved; this is called anoxygenic photosynthesis. Cyanobacteria are an exception: they use chlorophyll-a and water as donor and release oxygen, like plants. The light energy is trapped to form ATP and reducing power, which fix CO2 into organic compounds.

Question: 4. Explain the economic importance of bacteria.

Answer: Bacteria are economically very important. They are used in making curd, cheese, vinegar, alcohol and antibiotics such as streptomycin. Nitrogen-fixing bacteria increase soil fertility, and bacteria are used in leather tanning, retting of fibres, sewage treatment and biogas production. Some bacteria cause diseases, food spoilage and damage to crops, which is their harmful side. On the whole, their beneficial uses in industry, agriculture and medicine are enormous.

Question: 5. Explain the bacterial diseases in Man with their signs, symptoms, treatment and prevention.

Answer: Important bacterial diseases of man include cholera, typhoid, tuberculosis, pneumonia and tetanus. Cholera causes severe watery diarrhoea and dehydration and spreads through contaminated water; it is treated by oral rehydration and antibiotics and prevented by clean water

and hygiene. Typhoid causes prolonged fever and abdominal pain and is prevented by vaccination and safe food. Tuberculosis affects the lungs with chronic cough, fever and weight loss, and is treated with a full course of antibiotics. Cleanliness, vaccination and early treatment control these diseases.

Question: 6. Explain the bacterial diseases in plants with their signs, symptoms, treatment and prevention

Answer: Bacteria cause several plant diseases, such as citrus canker, bacterial blight of cotton and rice, and crown gall. Signs and symptoms include water-soaked spots, cankers, wilting, yellowing, rotting and abnormal growths on leaves, fruits and stems. Infected plants show reduced growth and yield. Treatment includes spraying bactericides and antibiotics and removing and burning infected plants. Prevention is by using disease-free seeds, crop rotation, resistant varieties and disinfection of tools, so the disease does not spread in the field.

Question: 7. What do you mean by bacterial growth? Describe its phases.

Answer: Bacterial growth means an increase in the number of bacterial cells by binary fission. When bacteria are grown in a culture, growth shows four phases. In the lag phase cells adjust to the medium and grow in size but do not divide much. In the log (exponential) phase they divide rapidly at a constant rate. In the stationary phase, nutrients decrease and waste increases, so birth and death rates become equal. In the decline (death) phase, cells die faster than they are produced.

3. Write detailed answers of the following questions:

Question: 1. What are the domains of classification? Differentiate them.

Answer: Living organisms are classified into three domains: Bacteria,

Archaea and Eukarya. Bacteria are true prokaryotes with peptidoglycan in their cell walls and are found everywhere. Archaea are also prokaryotic but their cell walls lack peptidoglycan, their membrane lipids are different, and many live in extreme environments like hot springs and salty lakes. Eukarya includes all eukaryotic organisms — protoctists, fungi, plants and animals — whose cells have a true nucleus and membrane-bound organelles. The domains differ in cell structure, biochemistry and habitat.

Question: 2. What are protobionts? Discuss their relationship [p with prokaryotes.

Answer: Protobionts were the first non-cellular structures formed from organic molecules on the early Earth, considered precursors of living cells. They could absorb substances, grow, and show simple metabolism and division, but they could not reproduce exactly like true cells. Over time, protobionts that developed a nucleic acid system and a membrane gave rise to the first true prokaryotic cells. Prokaryotes are therefore regarded as the evolved, fully cellular descendants of protobionts. This relationship explains the origin of the earliest bacteria-like organisms.

Question: 3. What are the extreme conditions Archea called?

Answer: Archaea living in extreme conditions are called extremophiles. Those living in very hot places, such as hot springs, are called thermophiles, and those tolerating both heat and acidity are thermoacidophiles. Archaea living in highly salty environments like salt lakes are called halophiles. Methanogens are archaea that live in oxygen-free places such as marshes and animal guts and produce methane gas. These archaea survive where most other organisms cannot live.

Question: 4. Differentiate between gram-positive and gram-negative bacteria.

Answer: Gram-positive bacteria have a thick peptidoglycan layer in their cell wall and no outer membrane; they retain the crystal violet stain and appear purple in Gram staining. Gram-negative bacteria have a thin peptidoglycan layer but possess an additional outer membrane containing lipopolysaccharides; they lose the violet stain and take the pink counter stain. Gram-negative bacteria are generally more resistant to antibiotics because of their outer membrane. Examples are Staphylococcus (gram-positive) and Escherichia coli (gram-negative).

Question: 5. Give the structure of peptidoglycan of bacterial cell wall.

Answer: Peptidoglycan is a large polymer that forms the rigid framework of the bacterial cell wall. It consists of long chains of two alternating sugars, N-acetylglucosamine (NAG) and N-acetylmuramic acid (NAM). Short peptide chains are attached to NAM units and cross-link the sugar chains together, forming a strong mesh-like network. This structure gives the wall strength, maintains cell shape and protects the bacterium from bursting. The enzyme lysozyme and the antibiotic penicillin attack this layer.

Question: 6. Differentiate between transformation and transduction in bacteria.

Answer: Transformation is the uptake of free DNA from the environment by a living bacterium, usually released from a dead bacterium, which changes its genetic characters. Transduction is the transfer of bacterial DNA from one bacterium to another through a bacteriophage (virus). In transformation no vector is involved, while in transduction a virus acts as the carrier. Both are methods of horizontal gene transfer and genetic recombination in bacteria, along with conjugation.

Question: 7. Why bacteria called recycling agent in nature?

Answer: Bacteria are called recycling agents because decomposer bacteria break down dead bodies of plants and animals and organic waste into simple inorganic substances. These substances, such as CO2, water, ammonia and minerals, return to the soil, air and water and are used again by plants. Without bacteria, dead matter would pile up and nutrients would remain locked in it. Bacteria also play a key role in nitrogen, carbon and sulphur cycles, keeping nature's recycling continuous.

Question: 8. How bacteria increase soil fertility? Give example.

Answer: Bacteria increase soil fertility mainly by fixing atmospheric nitrogen and decomposing organic matter. Rhizobium living in root nodules of leguminous plants converts nitrogen gas into ammonia compounds usable by plants. Free-living Azotobacter and cyanobacteria like Nostoc also fix nitrogen in soil. Decomposer bacteria convert dead remains into humus and release nitrates through nitrification. As a result the soil becomes rich in plant nutrients and crop yield increases.

Question: 9. How cholera and typhoid spread in human population?

Answer: Cholera spreads through water and food contaminated with faeces of an infected person containing Vibrio cholerae, especially during floods and poor sanitation. Houseflies can also transfer the germs to food. Typhoid spreads in a similar faecal-oral way through contaminated water, milk and food, and healthy carriers of Salmonella typhi can spread it unknowingly. Both diseases become epidemics where drinking water is unsafe and sewage disposal is poor. Boiling water and hygienic food handling prevent their spread.

Question: 10. What do we mean by normal flora of bacteria?

Answer: Normal flora means the microorganisms, mainly bacteria, that normally live on the skin and mucous membranes and inside the intestine of a healthy person without causing disease. The human body contains about one hundred trillion such bacteria. They prevent colonization by harmful pathogens by competing for attachment sites and nutrients. They also stimulate antibody production and some synthesize vitamins K and B12 in the large intestine. Internal tissues like blood and muscles are normally free of these microbes.

Question: 11. List down the physical and chemical methods to control bacteria.

Answer: Chemical methods include use of disinfectants and antiseptics such as phenol, halogens (chlorine and iodine), heavy metals, soaps and detergents, aldehydes, ethylene oxide, oxidizing agents and food preservatives, as well as antibiotics. Physical methods include boiling, dry-heat oven, incineration, autoclaving, pasteurization, refrigeration, freezing, high-pressure processing, desiccation, lyophilization, ionizing and non-ionizing radiation and membrane filtration. Sterilization by autoclave at 121°C for 15-40 minutes at 15 psi destroys even bacterial spores. The choice of method depends on the material to be treated.

Chapter 7: Protoctists and Fungi

1. Encircle the correct choice

(i) Which of the following is true about the Kingdom Protoctista

(a) All organisms are prokaryotes

✓ (b) All organisms are eukaryotes

(c) All organisms are autotrophic

(d) All organisms are heterotrophic

Reason: All protoctists are eukaryotes with a true nucleus, though some are autotrophic and others heterotrophic.

(ii) All members of green algae have the pigment combination

✓ (a) Chlorophylls a and b

(b) Chlorophylls a and c

(c) Chlorophyll a and Phycobilins

(d) Chlorophylls a , b and c

Reason: Green algae contain chlorophylls a and b, the same pigment combination as land plants.

(iii) The major grouping of protozoa is based upon their

(a) Feeding habits

(b) Mode of reproduction

(c) Mode of Nutrition

✓ (d) Mode of locomotion

Reason: Protozoa are grouped mainly on the basis of their mode of locomotion, e.g. pseudopodia, flagella, cilia or non-motile.

(iv) Protoctista having isomorphic alternation of generation

(a) Chlorella

✓ (b) Ulva

(c) Euglena

(d) Phytophthora

Reason: Ulva shows isomorphic alternation of generations, in which sporophyte and gametophyte look alike.

(v) The amoeboid stage of slime mold is

✓ (a) Plasmodium

(b) Fruiting bodies

(c) Euglena

(d) Merozoites

Reason: The amoeboid feeding stage of a slime mold is the multinucleate mass called plasmodium.

(vi) Trypanosomes and Leishmanias are disease causing

(a) Viruses

(b) Bacteria

✓ (c) Protoctists

(d) Fungi

Reason: Trypanosoma and Leishmania are disease-causing protoctists (flagellate protozoa) that cause sleeping sickness and kala-azar.

(vii) Ulva is commonly called

(a) Bacteria called

(b) Virus

(c) Mushroom

✓ (d) Sea lettuce

Reason: Ulva is commonly called sea lettuce because of its green

leaf-like thallus found in sea water.

(viii) The study of fungi is called

(a) Zoology

(b) Microbiology

(c) Botany

✓ (d) Mycology

Reason: The study of fungi is called mycology.

(ix) The most important cellulose degraders in ecosystem are

(a) Ascomycota

(b) Zygomycota

✓ (c) Basidiomycota

(d) Deutromycota

Reason: Basidiomycota are the most important cellulose (and wood) degraders in the ecosystem.

2. Write short answers of the following questions:

Question: 1. Why is the kingdom Protoctista considered to be an artificial taxon?

Answer: Protoctista is considered an artificial taxon because it is a polyphyletic group made of organisms that do not share a single common ancestor. It includes plant-like algae, animal-like protozoa and fungus-like slime molds, which are very different from each other. They are grouped together mainly because they do not fit into the plant, animal or fungus kingdoms. Biologists therefore regard this kingdom as an unnatural, convenience grouping.

Question: 2. Why we say that ulva has isomorphic alternation of generation?

Answer: Ulva has isomorphic alternation of generations because its two generations, the sporophyte and the gametophyte, are morphologically identical in shape and size. The diploid sporophyte produces haploid spores by meiosis, which grow into a similar-looking haploid gametophyte. The gametophyte produces gametes that fuse to restore the sporophyte. Since both generations look alike, the alternation is called isomorphic.

Question: 3. Why asexual and sexual ulva plants are called as sporophyte and gametophyte?

Answer: In Ulva, the plant that reproduces asexually by producing spores is called the sporophyte; it is diploid and forms spores by meiosis. The plant that reproduces sexually by producing gametes is called the gametophyte; it is haploid and its gametes fuse to form a zygote. The zygote grows into a new sporophyte. Thus the names are based on the type of reproductive cells each generation produces.

Question: 4. Protoctista are the link between prokaryotes and the more modern eukaryotic like plants and animals explain why?

Answer: Protoctista are regarded as a link because they are the earliest eukaryotes and resemble the ancestors of modern plants, animals and fungi. They show characteristics connecting simple prokaryotic organization with the advanced eukaryotic kingdoms. Plant-like protoctists resemble early plants, protozoa resemble early animals, and slime molds resemble early fungi. The three main eukaryotic kingdoms probably arose from protoctist ancestors. Hence they form an evolutionary bridge between prokaryotes and modern eukaryotes.

Question: 5. Name various classes of Protozoa and how each of their members move?

Answer: Protozoa are divided into classes based on locomotion. Rhizopoda (e.g. Amoeba) move by temporary pseudopodia, Flagellata (e.g. Euglena, Trypanosoma) move by whip-like flagella, and Ciliata (e.g. Paramecium) move by numerous small cilia. Sporozoa (e.g. Plasmodium) have no locomotory organelles in the adult stage and are mostly parasites. Each class is thus named after its organ of movement.

Question: 6. Describes some of the unusual features of nuclear behavior in fungi?

Answer: Fungi show unusual nuclear behaviour. Their nuclei are haploid for most of the life cycle, and the diploid phase is only the brief zygote stage. During mitosis the nuclear membrane does not break down and the spindle forms inside the nucleus, which is called nuclear (closed) mitosis. In many fungi, two haploid nuclei from different parents coexist in the same cell for a long time (dikaryotic condition) before fusing. Meiosis follows karyogamy immediately during sexual reproduction.

Question: 7. What features allow fungi to survive in all environment where life is possible?

Answer: Fungi survive almost everywhere because they produce enormous numbers of lightweight spores that spread by air and resist unfavourable conditions. Their hyphae can grow rapidly over and into food sources and absorb nutrients after external digestion. They can live as saprophytes, parasites or symbionts on a wide range of substrates. They also tolerate a wide range of temperature, pH and moisture, and can remain dormant as spores when conditions are harsh.

3. Write detailed answers of the following questions:

Question: 1. Write down general characteristics of Protoctista and diversity

among Protoctista?

Answer: Protoctista are eukaryotic organisms that do not belong to the plant, animal or fungal kingdoms. They may be unicellular, colonial or simple multicellular, and show autotrophic, heterotrophic, parasitic or absorptive nutrition. The group is very diverse: plant-like protoctists are the algae such as Chlorella and Ulva, animal-like protoctists are the protozoa such as Amoeba and Plasmodium, and fungus-like protoctists are slime molds and water molds like Phytophthora. They mostly live in water or moist habitats. Because of this diversity, the kingdom is considered polyphyletic.

Question: 2. Describe structure and reproduction in Ulva?

Answer: Ulva, or sea lettuce, has a flat, green, leaf-like thallus only two cells thick, attached to rocks by a holdfast. Its cells contain a cup-shaped chloroplast and it photosynthesizes like plants. Ulva reproduces both asexually and sexually with isomorphic alternation of generations. The diploid sporophyte produces haploid zoospores by meiosis, which grow into identical haploid gametophytes. Gametophytes release gametes that fuse to form a zygote, which develops into a new sporophyte, completing the cycle.

Question: 3. Give diagnostic features of four classes of fungi?

Answer: Zygomycota (e.g. Rhizopus, Mucor) have non-septate hyphae and form a thick-walled zygospore in sexual reproduction. Ascomycota (e.g. yeast, Aspergillus) form spores called ascospores inside a sac-like ascus and are called sac fungi. Basidiomycota (e.g. mushrooms, Agaricus) form basidiospores on a club-shaped basidium and have septate hyphae with a dikaryotic stage. Deuteromycota are the imperfect fungi in which sexual reproduction has not been observed; they reproduce only asexually by

conidia, e.g. Penicillium in some classifications.

Question: 4. Write an essay on economic importance of fungi?

Answer: Fungi are of great economic importance. Positively, yeasts are used in baking and brewing, fungi produce antibiotics like penicillin, and edible mushrooms are a source of food. Fungi are the main decomposers, recycling nutrients in nature, and mycorrhizal fungi improve plant growth; some fungi are used in biological control of pests and in biotechnology and genetic research. Negatively, fungi cause many diseases of crops (e.g. late blight, rusts and smuts) and of humans and animals, and they spoil stored food and damage wood and leather. Their benefits in medicine and industry, however, are immense.

Chapter 8: Diversity Among Plants

1. Encircle the correct choice

(i) The rapid diversification of plants occurred in the period:

(a) Silurian

✓ (b) Devonian

(c) Carboniferous

(d) Paleozoic

Reason: The rapid diversification of land (vascular) plants occurred in the Devonian period.

(ii) Spores of land plants tough, flexible and resistant to biodegradation by:

(a) Lignin

(b) Chitin

(c) Waxes

✓ (d) Sporopollenin

Reason: Sporopollenin makes spore and pollen walls tough, flexible and resistant to biodegradation.

(iii) Hepaticae, Anthocerotae and Moss are the classes of:

(a) Angiosperms

(b) Gymnosperms

(c) Tracheophytes

✓ (d) Bryophytes

Reason: Hepaticae (liverworts), Anthocerotae (hornworts) and Mosses are the classes of Bryophytes.

(iv) The most successful kind of reproduction is:

(a) Heterogamy

✓ (b) Oogamy

(c) Anisogamy

(d) Isogamy

Reason: Oogamy, with a large non-motile egg and small motile sperm, is the most successful kind of reproduction in land plants.

(v) The dominant generation of living vascular plants existing as free-living is:

✓ (a) Sporophyte

(b) Gametophyte

(c) Tracheophyte

(d) Pteridophyte

Reason: In living vascular plants the sporophyte is the dominant, free-living generation.

(vi) The most primitive living vascular plants is:

(a) Lycophyta

(b) Sphenophyta

✓ (c) Psilophyta

(d) Filicinophyta

Reason: Psilophyta (e.g. Psilotum) are the most primitive living vascular plants, lacking true roots and leaves.

(vii) The gametophyte of Fern is called:

(a) Archegonium

(b) Antheridium

(c) Cones

✓ (d) Prothallus

Reason: The free-living heart-shaped gametophyte of a fern is called the prothallus.

(viii) Double fertilization is the characteristic feature of:

(a) Bryophytes

(b) Gymnosperms

✓ (c) Angiosperms

(d) Ferns

Reason: Double fertilization is the characteristic feature of angiosperms (flowering plants).

(ix) Resins, turpentine and pine oil are obtained from:

(a) Rose

(b) Onion

✓ (c) Pins

(d) Banana

Reason: Resins, turpentine and pine oil are obtained from pines (Pins, i.e. Pinus).

(x) In banana tree, flowers are covered over by one or many large brackets called:

✓ (a) Spathes

(b) Spadix

(c) Capitulum

(d) Palea

Reason: In banana, the inflorescence/flowers are covered by one or many large bracts called spathes.

2. Write short answers of the following questions:

Question: 1. How life cycle of plants shows alternation of generations?

Answer: The plant life cycle alternates between a haploid gametophyte generation and a diploid sporophyte generation. The sporophyte produces haploid spores by meiosis, and the spores grow into the gametophyte. The gametophyte produces gametes, which fuse during fertilization to form a diploid zygote that grows into a new sporophyte. This regular alternation is found in all land plants, though the dominant generation differs among groups.

Question: 2. Why bryophytes are called non vascular plants?

Answer: Bryophytes are called non-vascular plants because they lack true vascular tissues, xylem and phloem, for conduction of water and food. They also lack true roots, stems and leaves, having instead rhizoids and simple leaf-like structures. Water moves in them slowly by diffusion from cell to cell. For this reason bryophytes remain small and live in moist places.

Question: 3. How plants cope when invaded the land from the sea?

Answer: When plants invaded land from the sea, they faced drying, temperature changes and lack of support. They coped by developing a waxy cuticle and stomata to control water loss, and spores with resistant sporopollenin walls. Vascular tissues and lignin gave conduction and mechanical support, and roots anchored them and absorbed water. Protected embryos and, later, seeds and pollen allowed reproduction without free water.

Question: 4. Why heterogamy is important for plants?

Answer: Heterogamy is important because the production of two different types of gametes allows specialization. The small motile male gamete can travel to the female gamete, while the large female gamete stores food for

the early embryo. This division of labour increases the chances of successful fertilization and survival of the offspring. Heterogamy reached its most successful form, oogamy, in land plants and led to seed formation.

Question: 5. Why flower is called a reproductive part of plant?

Answer: The flower is called the reproductive part of the plant because it bears the male and female reproductive organs. The stamen (anther and filament) produces pollen grains containing male gametes, and the carpel (stigma, style and ovary) contains ovules with the egg. Pollination and fertilization occur through the flower, after which the ovary forms fruit and ovules form seeds. Without flowers, sexual reproduction in angiosperms cannot take place.

Question: 6. Why fertilization in angiosperm is called double fertilization?

Answer: Fertilization in angiosperms is called double fertilization because two fusions occur inside the embryo sac. One male gamete fuses with the egg to form the diploid zygote (syngamy). The second male gamete fuses with the two polar nuclei to form a triploid primary endosperm nucleus, which develops into endosperm, the food tissue for the embryo. Since two fertilization events take place together, it is called double fertilization.

Question: 7. How seeds are evolved?

Answer: Seeds evolved from the heterospory of earlier plants, in which two types of spores, microspores and megaspores, were produced. The megaspore was retained inside the megasporangium (ovule) on the parent plant instead of being shed. A protective integument developed around the ovule, and pollen delivered the male gamete without needing water. After fertilization the ovule became the seed, containing an embryo with

stored food. This gave plants a great advantage for survival and dispersal on land.

Question: 8. Why gymnosperms have naked seeds but not angiosperms?
Answer: Gymnosperms have naked seeds because their ovules and seeds are not enclosed in an ovary; they lie exposed on cone scales. The word gymnosperm itself means 'naked seed'. In angiosperms, the ovules are enclosed inside the carpel's ovary, which after fertilization becomes the fruit wall. Therefore angiosperm seeds remain protected inside the fruit, while gymnosperm seeds stay naked.

3. Write detailed answers of the following questions:

Question: 1. Define plants. Discuss evolutionary relation in plants.
Answer: Plants are multicellular, photosynthetic eukaryotic organisms with cellulose cell walls, chlorophyll a and b, and a life cycle showing alternation of generations. Evolutionarily, land plants arose from green algal ancestors that invaded land. Bryophytes were the earliest land plants without vascular tissue, followed by vascular seedless plants like ferns. Gymnosperms then evolved naked seeds, and finally angiosperms evolved flowers, enclosed seeds and double fertilization, becoming the most successful group. Each step added adaptations for life on land.

Question: 2. Write down general characteristics of plants.
Answer: Plants are multicellular eukaryotes with cell walls of cellulose and chloroplasts containing chlorophylls a and b. They are mostly autotrophic and prepare food by photosynthesis, storing it as starch. Their life cycle shows alternation of a haploid gametophyte and a diploid sporophyte generation. Most plants are fixed in one place and show limited movement. They range from tiny mosses to giant trees and are adapted to

almost all habitats on Earth.

Question: 3. Describe lifecycle of Moss with diagram.

Answer: In moss, the dominant free-living generation is the haploid gametophyte, which grows from a spore through a filamentous protonema. The mature gametophyte bears antheridia producing male gametes and archegonia producing eggs; fertilization needs water and forms a diploid zygote. The zygote grows into a sporophyte (foot, seta and capsule) that remains attached to and dependent on the gametophyte. The capsule produces haploid spores by meiosis, which are dispersed by wind and germinate to repeat the cycle. (Diagram to be drawn as in the textbook.)

Question: 4. Why vascular plants are considered more successful land plants?

Answer: Vascular plants are considered more successful land plants because their xylem and phloem conduct water, minerals and food efficiently over long distances. Lignified xylem also gives mechanical strength, allowing them to grow tall and compete for light. They have true roots, stems and leaves, a dominant sporophyte, and in seed plants, pollen and seeds that free reproduction from water. These adaptations let them inhabit diverse and dry habitats and dominate the land flora.

Question: 5. How leaves are evolved?

Answer: Leaves evolved in two main ways during plant evolution. Microphylls, as in Lycophyta, developed as small outgrowths (enations) of the stem that later received a single vascular strand. Megaphylls, found in ferns and seed plants, evolved from flattened and webbed branching systems: side branches became flat, arranged in one plane, and the

spaces between them filled with photosynthetic tissue. Megaphylls have a branched vein system and a large surface for photosynthesis. This evolution greatly increased the photosynthetic efficiency of land plants.

Chapter 9: Diversity Among Animals

1. Encircle the correct choice.

(i) Choose the incorrect pair?

(a) Sponges → Spicules.

(b) Cnidaria → Nematocyts.

✓ (c) Segmented worm → Psuedopodia.

(d) Arthropoda → jointed legs.

Reason: Segmented worms (annelids) move by setae/parapodia, not pseudopodia; pseudopodia belong to Amoeba, so this pair is incorrect.

(ii) Water movement through sponge?

✓ (a) Ostia → Spongocoel → Osculum.

(b) Spongocoel → Osculum → Ostia.

(c) Ostia→ Osculum → Spongocoel.

(d) Osculum → Spongocoel → Ostia.

Reason: Water enters the sponge through ostia, passes into the spongocoel and leaves through the osculum.

(iii) Choose the term which encompasses all other in the list

(a) Coelomate

(b) Protostome

✓ (c) Bilateria

(d) Triploblastic

Reason: Bilateria encompasses all the others, as coelomates, protostomes and triploblastic animals are all subgroups of bilaterally symmetrical animals.

(iv) Dry scaly skin, ectothermic, two pairs of legs, lungs, Internal Fertilization, eggs with amniotic shell, and a three chambered heart are all properties of which class?

(a) Amphibia

✓ (b) Reptilia

(c) Aves

(d) Mammals

Reason: These are the properties of class Reptilia.

(v) Pick the odd one out?

(a) Star fish.

(b) Brittle star.

✓ (c) Lamprey.

(d) Sea urchin.

Reason: Star fish, brittle star and sea urchin are echinoderms, while lamprey is a vertebrate (jawless fish), so it is the odd one out.

(vi) Which of the following terms or structures is properly associated only with animals?

✓ (a) Genes.

(b) Cell wall.

(c) Autotrophy.

(d) Sexual reproduction.

Reason: Among the options, the (Hox) genes that control animal body patterning are properly associated only with animals; cell wall and autotrophy are absent in animals and sexual reproduction also occurs in plants and fungi.

(vii) An adult animal that possesses bilateral symmetry is most certainly

also

(a) Triploblastic.

(b) Deuterostomes.

(c) Coelomate.

(d) Protostomes.

Reason: All bilaterally symmetrical adult animals are triploblastic, having three germ layers, though they may or may not be coelomate, protostome or deuterostome.

(viii) Protostome characteristics generally include which of the following?

(a) A Mouth that develops secondly and far away from the blastopore

(b) Radial body

(c) Radial cleavage.

(d) Determinate cleavage

Reason: Protostomes generally show spiral and determinate cleavage, and the mouth develops from or near the blastopore.

(ix) Sponges structural materials (spicules) are manufactured by the

(a) Pinacocytes

(b) Choanocytes.

(c) Amoebocytes.

(d) Porocytes.

Reason: Amoebocytes (scleroblasts) manufacture the spicules that form the skeleton/structural material of sponges.

(x) Which of the following can be used to distinguish a nematode from annelids? I. Type of body cavity II. Number of muscle layers in the body wall III. Presence of segmentation IV. Number of embryonic tissue layers

(a) I only

(b) II only

(c) I and II

(d) I, II and III

Reason: Nematodes differ from annelids in type of body cavity (pseudocoelom vs true coelom), number of muscle layers in the body wall, and presence of segmentation (absent in nematodes, present in annelids).

2. Write short answers of the following questions:

Question: 1. How animals are classified?

Answer: Animals are classified into about 33 phyla on the basis of their evolutionary relationships and body organization. The criteria include level of organization, number of tissue (germ) layers, symmetry, presence or absence and type of coelom, and the fate of the blastopore. Only nine major phyla, from Porifera to Chordata, are usually studied in depth, while the rest are minor phyla. Comparative morphology and embryological patterns provide the main clues for this classification.

Question: 2. Why sponges placed in animal kingdom, when it lacks tissue organization?

Answer: Sponges are placed in the animal kingdom because they are multicellular, heterotrophic eukaryotes without cell walls, like other animals. They ingest food particles, have animal-like cells such as choanocytes, and reproduce sexually producing a free-swimming larva. Although they have only a cellular level of organization and lack true tissues, their nutrition, structure and development clearly show animal affinities. Hence they are considered the simplest animals, in phylum Porifera.

Question: 3. Why flat worms are named so?

Answer: Flatworms (phylum Platyhelminthes) are named so because their bodies are dorsoventrally flattened like a ribbon or leaf. This flat shape keeps every cell close to the surface, so gases and food can diffuse easily, as they have no circulatory system. They are triploblastic, bilaterally symmetrical and acoelomate animals. Examples are Planaria, tapeworm and liver fluke.

Question: 4. Differentiate between invertebrate chordates and vertebrates

Answer: Invertebrate chordates possess a notochord but no vertebral column or backbone; examples are Amphioxus, tunicates and hagfish-like forms in the subphyla Urochordata and Cephalochordata. Vertebrates have the notochord replaced by a vertebral column made of vertebrae, a well-developed skull and brain, and an internal skeleton. Invertebrate chordates are small and mostly marine, while vertebrates include fishes, amphibians, reptiles, birds and mammals. Both groups, however, show chordate characters at some stage of life.

Question: 5. Differentiate between cartilaginous fishes and bony fishes

Answer: Cartilaginous fishes (e.g. sharks, rays) have a skeleton made of cartilage, placoid scales, a heterocercal tail and no swim bladder or operculum. Bony fishes (e.g. rohu, trout) have a skeleton mainly of bone, cycloid or ctenoid scales, a homocercal tail and a gill cover (operculum). Most cartilaginous fishes are marine, while bony fishes live in both marine and fresh water. Fertilization and other habits also differ between the two classes.

Question: 6. Why protostomes are named so?

Answer: Protostomes are named so because in their embryonic

development the blastopore, the first opening of the embryo, develops into the mouth. The word protostome means 'mouth first'. The anus is formed later at a separate place. Molluscs, annelids and arthropods belong to this group. In contrast, in deuterostomes the blastopore becomes the anus and the mouth is formed secondarily.

Question: 7. Why diploblastic animals are different from the triploblastic?

Answer: Diploblastic animals, such as cnidarians, develop from only two germ layers, ectoderm and endoderm, with a jelly-like mesoglea between them. Triploblastic animals develop from three germ layers, ectoderm, mesoderm and endoderm. The mesoderm allows triploblastic animals to form complex organs, muscles and, in many, a coelom. Therefore triploblastic animals show a much higher level of organization than diploblastic animals.

Question: 8. Why echinoderms are placed in bilateria when adults are radially symmetrical?

Answer: Echinoderms are placed in Bilateria because their larvae are bilaterally symmetrical, showing their true evolutionary relationship. The radial symmetry of adults is a secondary adaptation to their slow-moving or sessile bottom-dwelling life. Their development is deuterostome-like, similar to chordates, which also supports this placement. Adult radial symmetry therefore does not reflect their ancestry.

3. Write detailed answers of the following questions:

Question: 1. Discuss the basic factors which helps in classification of animals

Answer: Animal classification is based mainly on level of organization (cellular, tissue, organ or organ-system level), number of germ layers

(diploblastic or triploblastic), and symmetry (asymmetrical, radial or bilateral). Another basic factor is the coelom: animals may be acoelomate, pseudocoelomate or true coelomate. The fate of the blastopore divides triploblastic coelomates into protostomes and deuterostomes, and the presence of a notochord separates chordates from invertebrates. Comparative morphology and embryological development provide evidence of evolutionary relationships used in this classification.

Question: 2. Write down evolutionary adaptations of phylum annelida.
Answer: Annelids show important evolutionary advances over simpler worms. Their body is metamerically segmented, with repeating units that allow specialization of different body regions. They possess a true coelom that acts as a hydrostatic skeleton and permits efficient burrowing movement. A closed circulatory system, complete digestive tract, and well-developed nervous and excretory systems (nephridia in each segment) improve their physiology. Setae and, in polychaetes, parapodia help in locomotion, making annelids successful in marine, freshwater and moist land habitats.

Question: 3. Write down the general characters of phylum arthropoda
Answer: Arthropods are triploblastic, bilaterally symmetrical, coelomate animals with a segmented body divided into head, thorax and abdomen (or cephalothorax and abdomen). They have jointed appendages, from which the phylum gets its name, and a hard exoskeleton of chitin that is moulted during growth. They have a complete digestive system, open circulatory system, and respiration by gills, tracheae or book lungs. Arthropods are the largest phylum, including insects, spiders, crustaceans and myriapods, adapted to every habitat.

Question: 4. Describe the general characters of phylum mollusca

Answer: Molluscs are soft-bodied, unsegmented, triploblastic coelomate animals, usually protected by a shell of calcium carbonate secreted by the mantle. Their body is divided into a head, a muscular ventral foot used for locomotion, and a visceral mass containing the organs. A radula in the mouth is a unique rasping feeding organ in most molluscs. They respire by gills or mantle cavity and mostly have an open circulatory system. Examples are snails, mussels, octopus and squid.

Question: 5. What are mammals? Name its sub-classes. Give the characters and examples of each sub class.

Answer: Mammals are warm-blooded vertebrates that feed their young on milk from mammary glands, have hair or fur on the body, and possess a diaphragm and a four-chambered heart. Subclass Prototheria (monotremes) lay eggs, have mammary glands without nipples and a cloaca; e.g. echidna and duck-billed platypus. Subclass Metatheria (marsupials) give birth to tiny immature young that complete development in a mother's pouch (marsupium); e.g. kangaroo and koala. Subclass Eutheria are placental mammals whose young develop fully inside the uterus nourished through a placenta; e.g. human, whale, bat and lion.

Chapter 10: Forms and Functions in Plant

1. Encircle the correct choice

(i) From where Plants primarily obtain essential nutrients:

(a) Water

(b) Air

✓ (c) Soil

(d) Light

Reason: Plants primarily obtain their essential mineral nutrients from the soil through their roots.

(ii) Identify a micro nutrient for plants amongst the following:

(a) Potassium

(b) Phosphorus

✓ (c) Iron

(d) Sluphur

Reason: Iron is a micronutrient, needed in trace amounts, while potassium, phosphorus and sulphur are macronutrients.

(iii) In which mineral deficient soil carnivorous plant typically live:

✓ (a) Nitrogen

(b) Calcium

(c) Magnesium

(d) Potassium

Reason: Carnivorous plants typically live in nitrogen-deficient soils and obtain nitrogen by digesting insects.

(iv) The plants growing in salt marshes close to sea are termed as:

(a) Hydrophytes

(b) Xerophytes

(c) Mesophytes

✓ (d) Halophytes

Reason: Plants growing in salt marshes close to the sea are called halophytes.

(v) The cause of color of leaf turns in to dull or bluish green is the deficiency of:

(a) Nitrogen

✓ (b) Phosphate

(c) Potassium

(d) Magnesium

Reason: Deficiency of phosphate causes leaves to turn dull or bluish green (often with purplish tint) and growth is stunted.

(vi) Which principle type of auxin of higher plants synthesized at the apices of stem and root:

(a) PAA

(b) 24-D

✓ (c) IAA

(d) NAA

Reason: IAA (indole acetic acid) is the principal natural auxin of higher plants, synthesized at the apices of stem and root.

(vii) Active transport of sucrose into sieve elements is a step commonly called:

✓ (a) Phloem loading

(b) unloading

(c) Diffusion

(d) Osmosis

Reason: Active transport of sucrose into sieve elements at the source is called phloem loading.

(viii) The capacity of a living system to lose water is:

(a) Osmotic pressure

✓ (b) Water potential

(c) Osmosis

(d) Plasma

Reason: Water potential describes the capacity/tendency of a living system to lose (or gain) water.

(ix) The production of lateral appendages depends on the activity of:

(a) Lateral meristem

(b) Iutercalary meristem

✓ (c) Apical meristem

(d) none of these

Reason: The production of lateral appendages (leaves and branches) depends on the activity of the apical meristem.

(x) Which process involved in the promotion of flowering by cold treatment:

(a) Photoperiodism

✓ (b) Vernalization

(c) Secondary Growth

(d) Transpiration

Reason: Promotion of flowering by cold treatment is called vernalization.

2. Write short answers of the following questions:

Question: 1. Why mineral nutrients are necessary for plants?

Answer: Mineral nutrients are necessary because plants use them to build essential compounds and to carry out metabolism. Nitrogen is needed for proteins and chlorophyll, phosphorus for ATP, nucleic acids and membranes, and potassium for enzyme activation and stomatal working. Without these minerals, plants show deficiency symptoms, poor growth and low yield. Although needed in different amounts, each essential element has a specific role that no other element can fully replace.

Question: 2. Why N2 is included in mineral nutrient although it is not mineral?

Answer: Nitrogen is included among mineral nutrients although it is not itself a mineral obtained as a metal, because plants absorb it from the soil in mineral (inorganic) forms, as nitrates and ammonium ions. It is an essential macronutrient required in large amounts for amino acids, proteins, nucleic acids and chlorophyll. Atmospheric N2 cannot be used directly by most plants and must first be fixed into these mineral forms. For this reason nitrogen is listed with the mineral nutrients.

Question: 3. Why desert plants reduce their leaf size?

Answer: Desert plants reduce their leaf size to decrease the surface area available for transpiration, so that water loss is minimized in hot and dry conditions. Small leaves, or leaves changed into spines as in cactus, also reduce heating of the plant. Photosynthesis is then often carried out by the green stem. This adaptation, together with thick cuticle and sunken stomata, helps desert plants conserve water.

Question: 4. Why carnivores plants use insect as food?

Answer: Carnivorous plants grow in soils that are deficient in nitrogen and other minerals, so they cannot get enough nitrogen from their roots. They trap and digest insects to obtain nitrogen and other nutrients from insect proteins. Their leaves are modified into traps like pitchers and snap traps with digestive enzymes. They still make their own food by photosynthesis; insects only supplement their mineral nutrition.

Question: 5. Why phytochrome pigments are important to photoperiodism?

Answer: Phytochrome is the light-sensitive pigment that enables plants to measure day and night length, which is the basis of photoperiodism. It exists in two interconvertible forms, Pr and Pfr, which change with red and far-red light. The amount of active Pfr at the end of the night informs the plant whether the critical night length has passed. Accordingly, short-day and long-day plants initiate flowering at the proper season.

Question: 6. How annual rings are formed?

Answer: Annual rings are formed by the seasonal activity of the vascular cambium during secondary growth. In spring, cambium produces large, thin-walled xylem cells (early wood), while in late summer and autumn it produces smaller, thick-walled cells (late wood). The sharp contrast between late wood of one year and early wood of the next appears as a ring. Counting these rings gives the approximate age of the tree.

Question: 7. Why cold treatment is necessary for the germination of seed in some plants?

Answer: In some plants, seeds remain dormant even when water and oxygen are available, because growth inhibitors are present or the embryo is not fully ready. Exposure to a period of low temperature (cold

treatment/stratification) breaks this dormancy by changing the balance of hormones, decreasing abscisic acid and increasing gibberellins. This ensures that seeds germinate in spring, not in the unfavourable winter. Hence cold treatment is necessary for germination of such seeds.

Question: 8. How osmotic adjustment is beneficial for plants?

Answer: In osmotic adjustment, plant cells accumulate solutes such as salts, sugars and amino acids in their vacuoles and cytoplasm. This lowers the cell's osmotic (water) potential, so the plant can draw and retain water even when soil water is scarce or salty. It helps maintain turgor, keeps stomata functioning and allows growth to continue during drought. Plants showing good osmotic adjustment are therefore more tolerant of water stress and salinity.

3. Write detailed answers of the following questions:

Question: 1. Give an account of exchange of gases in plants?

Answer: Gaseous exchange in plants occurs mainly through stomata in leaves and lenticels in woody stems, and through the general surface of roots. During the day, CO_2 enters through open stomata for photosynthesis and oxygen produced is released. For respiration, oxygen diffuses in and CO_2 diffuses out, day and night, according to concentration gradients. Guard cells regulate stomatal opening to balance CO_2 uptake with water loss by transpiration. Intercellular spaces inside the leaf connect the stomata to all photosynthetic cells, making exchange efficient.

Question: 2. How does pressure flow theory explain the movement of sugar through the plants?

Answer: According to the pressure flow theory, sucrose is actively loaded

into sieve tubes at the source (leaf), lowering their water potential. Water enters by osmosis from the xylem, creating high turgor pressure that pushes the sap towards the sink (root, fruit or growing part). At the sink, sucrose is unloaded and used or stored, water leaves the sieve tubes, and pressure falls. This pressure difference between source and sink drives the mass flow of sugars through the phloem.

Question: 3. Describe the cohesion tension theory of water movement through xylem? What supplies the cohesion and what is the source of tension? How do these two interact to move water through a plant?
Answer: The cohesion-tension theory explains the ascent of sap in xylem. Transpiration from leaves creates a tension (negative pressure) at the top, which is the source of the pulling force. Water molecules cohere to each other by hydrogen bonds, forming a continuous unbroken column in the xylem, and also adhere to xylem walls. The transpiration pull is transmitted down this cohesive column, drawing water from roots to the tallest leaves. Thus cohesion keeps the column intact while tension pulls it upward.

Question: 4. What essential inorganic nutrient does a plant require for growth, which of these nutrients are required in trace amounts? What are some of the general functions of inorganic nutrients?
Answer: Plants require essential inorganic nutrients divided into macronutrients, needed in large amounts — nitrogen, phosphorus, potassium, sulphur, calcium, magnesium (and sodium, chlorine in the book's list) — and micronutrients needed in trace amounts, such as iron, manganese, boron, zinc, copper, molybdenum and cobalt. In general, inorganic nutrients form structural parts of proteins, chlorophyll, nucleic acids and membranes. They also act as enzyme activators and cofactors,

maintain osmotic balance and pH, and take part in energy transfer, e.g. phosphorus in ATP. Deficiency of any essential nutrient disturbs growth.

Question: 5. What are growth regulators? Name and discuss five in detail?

Answer: Growth regulators (plant hormones) are organic substances produced in very small amounts that control growth and development. Auxins (e.g. IAA) promote cell elongation, root initiation and apical dominance. Gibberellins promote stem elongation, seed germination and flowering. Cytokinins promote cell division and delay leaf ageing, working with auxins in tissue growth. Abscisic acid inhibits growth, promotes dormancy and closes stomata during water stress. Ethene (ethylene) promotes fruit ripening, leaf fall and senescence. Together these five regulators coordinate the plant's life activities.

Question: 6. Discuss categorization of plants based on osmotic adjustment?

Answer: On the basis of water relations and osmotic adjustment, plants are categorized as hydrophytes, mesophytes, xerophytes and halophytes. Hydrophytes live in water, have little need for osmotic adjustment, thin cuticle and large air spaces. Mesophytes live in moderate water conditions and show ordinary regulation of stomata and turgor. Xerophytes of deserts adjust osmotically by accumulating solutes, reducing leaves and storing water to survive drought. Halophytes of salty marshes accumulate salts in vacuoles and may excrete excess salt, maintaining water uptake from saline soil.

Chapter 11: Holozoic Nutrition

1. Encircle the correct choice.

(i) The process of nutrition does not include this:

(a) Taking in of food

(b) Breaking down of food

(c) Absorption of nutrients

✓ (d) Removal of undigested food

Reason: Removal of undigested food is egestion, a separate elimination step; nutrition proper consists of ingestion, digestion, absorption and assimilation.

(ii) The chemicals playing the most important role in digestion of food are:

(a) Hormones

✓ (b) Enzymes

(c) HCl

Reason: Enzymes play the most important role, as they chemically break down complex food into simple soluble molecules during digestion.

(iii) The thecodont condition refers to:

(a) Presence of two sets of teeth during life time

(b) all alike teeth

(c) Sharp, pointed teeth

✓ (d) Embedded teeth

Reason: Thecodont means teeth embedded in sockets in the jaw bone, as in mammals and crocodiles.

(iv) The type of digestion in Planaria is:

(a) Intracellular

(b) Extracellular

(c) Intercellular

✓ (d) Both a & b

Reason: Planaria shows both extracellular digestion in the gut cavity and intracellular digestion inside food vacuoles of gut cells.

(v) Number of milk teeth is:

(a) 10

(b) 15

✓ (c) 20

(d) 25

Reason: A child has 20 milk (deciduous) teeth, which are later replaced by permanent teeth.

(vi) Teeth meant for cutting are:

✓ (a) Incisors

(b) Canines

(c) Pre-molars

(d) Molars

Reason: Incisors are sharp, chisel-shaped teeth meant for cutting and biting food.

(vii) The part of gut also associated with respiratory system is:

(a) Stomach

(b) Esophagus

✓ (c) Pharynx

(d) Both b & c

Reason: The pharynx is a common passage for food and air, so it is associated with both the gut and the respiratory system.

(viii) The number of salivary glands in our oral cavity are:

(a) 3

(b) 4

(c) 5

✓ (d) 6

Reason: There are six salivary glands, i.e. three pairs (parotid, submaxillary/submandibular and sublingual), opening into the oral cavity.

(ix) The rugae are related to:

✓ (a) Stomach

(b) Duodenum

(c) Jejunum

(d) Ilium

Reason: Rugae are the folds of the inner wall of the stomach that allow it to expand and help in churning food.

(x) BMI over 30 kg/m² is considered as:

(a) Normal

(b) Over-weight

(c) under-weight

✓ (d) Obesity

Reason: According to WHO, a body mass index over 30 kg/m² in an adult is considered obesity.

2. Write short answers of the following questions:

Question: 1. List out the stages in holozoic nutrition.

Answer: Holozoic nutrition consists of five stages. Ingestion is the taking in of food through the mouth, digestion is the mechanical and chemical breakdown of complex food into simple soluble molecules, and absorption is the uptake of these molecules through the intestinal lining into blood. Assimilation is the use of absorbed food for energy, growth and repair, and egestion is the removal of undigested food from the body. These stages occur in sequence in animals like man.

Question: 2. How stomach itself is protected against the strong HCl?

Answer: The stomach is protected from its own strong HCl and pepsin by a thick layer of mucus secreted by its inner wall, which forms a physical barrier. The epithelial cells of the stomach lining are tightly joined and are replaced rapidly when damaged. Pepsin is secreted in an inactive form, pepsinogen, and is activated only in the lumen. Bicarbonate in the mucus also neutralizes acid near the wall. Together these protections prevent autodigestion of the stomach.

Question: 3. Point out the ways through which the internal surface area of the small intestine is increased.

Answer: The internal surface area of the small intestine is greatly increased in three ways. The inner wall has circular folds, the surface bears millions of finger-like villi, and each epithelial cell of a villus has tiny microvilli forming a brush border. These structures multiply the absorptive area many times. A large surface area, thin walls and a rich blood and lymph supply make absorption of digested food very efficient.

Question: 4. How bile help in the digestion of fats?

Answer: Bile, produced by the liver and stored in the gall bladder, contains bile salts but no enzymes. Bile salts break large fat globules into tiny

droplets, a process called emulsification, which greatly increases the surface area of fats. This allows the enzyme lipase to digest fats quickly into fatty acids and glycerol. Bile also neutralizes the acidic chyme from the stomach, providing the alkaline medium needed by intestinal enzymes.

Question: 5. Enlist the role of large intestine.

Answer: The large intestine mainly absorbs water and salts from the liquid material received from the small intestine, converting it into semi-solid faeces. It also absorbs some vitamins produced by its bacteria, such as vitamin K and B vitamins. Its mucus lubricates the passage of faeces. Finally, it stores faeces temporarily and removes them by egestion through the anus. It has no role in enzymatic digestion of food.

Question: 6. What accounts for the presence of bacteria in large intestine?

Answer: Bacteria reach the large intestine with food and water and find there a warm environment, moisture and a continuous supply of undigested material on which to feed. Unlike the stomach, the large intestine has no strong acid to kill them, so they multiply and live as a normal flora. These bacteria are mostly harmless or useful; they produce vitamins and gases and help form faeces. Their presence is therefore a normal feature of the gut.

Question: 7. What are the health risks involved in obesity?

Answer: Obesity increases the risk of many serious diseases. It is linked with type 2 diabetes, high blood pressure, heart and cardiovascular diseases, stroke and certain cancers. Excess weight also strains joints, causing arthritis, and can disturb breathing during sleep. According to the book, obesity results from inheritance, environment, diet and lack of

exercise combined. Long-term proper diet and regular physical activity are needed to control it.

Question: 8. List out some factors which may lead to obesity.

Answer: Obesity develops from a combination of factors. Genetic inheritance can make a person prone to storing fat, while an environment with abundant energy-rich food and little physical work encourages weight gain. Eating excess fats and sugars, overeating, and a sedentary lifestyle are major causes. Hormonal disorders and certain medications may also contribute. When energy intake remains greater than energy expenditure for a long time, fat accumulates and obesity results.

Question: 9. What is Anorexia nervosa?

Answer: Anorexia nervosa is a psychological eating disorder in which a person has an intense fear of gaining weight and refuses to eat adequately. It is most common in young girls and women. The victim loses appetite and weight falls well below normal; some patients also practise induced vomiting. It can cause severe weakness, hormonal disturbance and even death if untreated. Treatment needs psychological help, medical care and proper nutrition.

3. Write detailed answers of the following questions:

Question: 1. State and explain the process of digestion in Amoeba?

Answer: Amoeba shows intracellular digestion. When Amoeba contacts a food particle, it extends pseudopodia around it and engulfs it by phagocytosis, forming a food vacuole. Lysosomes fuse with the vacuole and pour digestive enzymes into it, first making it acidic and then alkaline. The food is digested inside the vacuole and the soluble products are absorbed into the surrounding cytoplasm and assimilated. The undigested

residue is removed by egestion when the vacuole reaches the cell surface and opens out.

Question: 2. Categorize the animals based upon their size of the food? Give examples of each group.

Answer: On the basis of the size of food they take, animals are categorized into microphagous, macrophagous and liquid feeders. Microphagous animals feed on tiny particles, e.g. Amoeba, Paramecium and many filter-feeding invertebrates. Macrophagous animals take in large pieces of food or whole organisms, e.g. Hydra, frog and man. Liquid feeders suck fluids, e.g. mosquitoes and leeches suck blood, butterflies suck nectar and spiders suck liquefied prey. This grouping reflects different methods of ingestion in holozoic animals.

Question: 3. Discuss the process of digestion in stomach?

Answer: In the stomach, food is churned by muscular contractions and mixed with gastric juice to form a semi-liquid chyme. Gastric juice contains HCl, which kills germs, provides an acidic medium and converts inactive pepsinogen into pepsin. Pepsin begins protein digestion, breaking proteins into peptones, while in infants rennin curdles milk protein. The stomach wall is protected by mucus, and partly digested food is gradually released into the duodenum through the pyloric sphincter. Stomach digestion is mainly of proteins.

Question: 4. State and explain the role of accessory glands associated with our gut.

Answer: The accessory glands of the human gut are the salivary glands, liver and pancreas, along with glands in the stomach and intestinal walls. Salivary glands secrete saliva containing amylase that starts starch

digestion and moistens food. The liver produces bile, which emulsifies fats, and also processes absorbed nutrients carried by the hepatic portal vein; the pancreas secretes pancreatic juice with enzymes for proteins, fats and carbohydrates and also insulin. Gastric and intestinal glands add their own juices. Together these glands make complete chemical digestion possible.

Question: 5. Explain the process of digestion and absorption of food in small intestine.

Answer: In the small intestine, chyme is mixed with bile, pancreatic juice and intestinal juice in the duodenum and jejunum, and digestion of all food types is completed here. Pancreatic amylase, lipase and proteases and intestinal enzymes convert carbohydrates to glucose, fats to fatty acids and glycerol, and proteins to amino acids. The products are absorbed through the villi: glucose, amino acids, vitamins and salts enter blood capillaries, while fatty acids and glycerol enter the lacteal (lymph) of the villus. The enormous surface provided by folds, villi and microvilli makes absorption highly efficient.

Question: 6. State and explain the sac-like gut with suitable example.

Answer: A sac-like gut is an incomplete digestive system with only one opening that serves as both mouth and anus. Food enters through the mouth into a sac-like gastrovascular cavity, where extracellular and intracellular digestion take place. Undigested residue is egested through the same opening, so feeding and egestion cannot occur at the same time. Hydra and Planaria are suitable examples of animals with a sac-like gut. It is simpler than the tube-like gut of higher animals, which has separate mouth and anus.

Chapter 12: Circulation

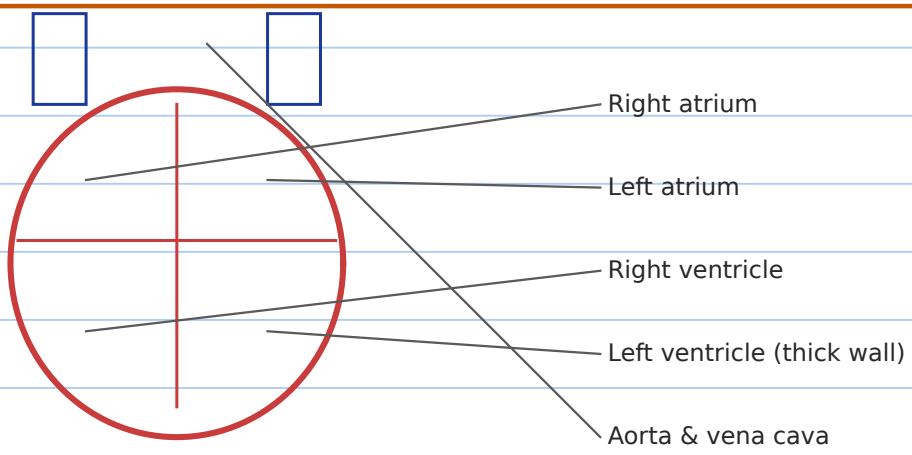


Fig: The Human Heart (Schematic)

1. Encircle the correct choice.

(i) Lymph most closely resembles which of the following?

(a) Blood

(b) Urine

(c) Plasma

✓ (d) Interstitial fluid

Reason: Lymph is essentially interstitial (tissue) fluid collected into lymph vessels, so it most closely resembles interstitial fluid.

(ii) What event initiates blood clotting?

✓ (a) Contact with an irregular surface by platelets and other factors in plasma.

(b) Production of the enzyme thrombin

(c) Conversion of fibrinogen into fibrin

(d) Excess flow of blood through a capillary

Reason: Clotting is initiated when platelets and plasma factors come in contact with an irregular (damaged/injured) surface.

(iii) Damage to the S.A Node in humans

(a) Is a major contributor to heart attacks.

(b) Would block conductance between the bundle branches and the purkinje fibers.

(c) Would have a negative effect on peripheral resistance

✓ (d) Would disrupt the rate and timing of cardiac muscle contractions.

Reason: The SA node is the pacemaker, so its damage would disrupt the rate and timing of cardiac muscle contractions.

(iv) Which of the following is measured by an electrocardiogram?

(a) Impulses from the AV node

(b) Impulses of the parasympathetic nervous system that control heartbeat

✓ (c) The speed on impulses from the SA node

(d) Contraction of the two atria

Reason: An electrocardiogram records the electrical impulses of the heart originating from the SA node and spreading through the heart muscle.

(v) Where are semilunar valves to be found in the mammalian heart?

(a) Where blood goes from atria to ventricles

(b) On the right side of the heart only

(c) Where the pulmonary veins attach to the heart

✓ (d) At the places where blood leaves via the aorta and pulmonary arteries

Reason: Semilunar valves are located where blood leaves the ventricles through the aorta and the pulmonary artery.

(vi) Organisms in which a circulating body fluid is distinct from the fluid

that directly surrounds the body's cells are likely to have which of the following?

(a) An open circulatory system

✓ (b) A closed circulatory system

(c) A gastrovascular cavity

(d) Branched tracheae

Reason: In a closed circulatory system blood is confined to vessels and remains distinct from the interstitial fluid surrounding the cells.

(vii) Average blood pressure is lowest in which structure (s)

(a) Aorta

(b) Arterioles

(c) Capillaries

✓ (d) Vena cava

Reason: Blood pressure falls continuously as blood travels away from the heart and is lowest in the vena cava returning blood to the heart.

(viii) Which of the following is/ are a cause (s) of vasoconstriction?

(a) Lying down after standing

(b) Standing after lying down

✓ (c) Stress or hormone concentration

(d) Histamine secretion

Reason: Stress and certain hormones (e.g. adrenaline) cause vasoconstriction, narrowing the blood vessels.

(ix) Blood returning to the human heart in a pulmonary vein drains first into the?

✓ (a) Left atrium

(b) Right atrium

(c) Left ventricle

(d) Right ventricle

Reason: Oxygenated blood returning through the pulmonary vein drains first into the left atrium.

(x) Which of the following is NOT an important function of the human circulation?

(a) Transport of nutrients and respiratory gases

(b) Regulation of body temperature

(c) Protection of the body by circulating antibodies



(d) Removal of waste products for excretion from the body

Reason: Circulation only transports waste products to the excretory organs; the actual removal/excretion from the body is performed by the kidneys etc., not by circulation itself. (Note: transport of wastes towards excretion is done by blood, but excretion itself is not a function of circulation.)

2. Write short answers of the following questions:

Question: 1. Why higher animals need circulatory system?

Answer: Higher animals are large and multicellular, so simple diffusion cannot supply materials to all cells quickly enough. Their cells lie far from the body surface and from the gut, and they have a high rate of metabolism. A circulatory system transports oxygen, nutrients, hormones and waste products rapidly to and from every cell through blood vessels. It also helps in temperature regulation and defence, so higher animals cannot survive without it.

Question: 2. Why SA node is called pace maker of heart?

Answer: The SA (sinoatrial) node is called the pacemaker because it

initiates each heartbeat by generating electrical impulses at a regular rhythm. These impulses spread over the atria, making them contract, and then reach the AV node and ventricles. The SA node sets the normal rate and timing of the heart's pumping, about 70-75 times per minute at rest. If it is damaged, the rhythm of the heart is disturbed.

Question: 3. Why the capillaries have a single layer of endothelium?

Answer: Capillaries have walls of a single layer of endothelium so that exchange of materials can occur easily and rapidly. Oxygen, nutrients, hormones and waste products diffuse across this very thin wall between blood and tissue cells. A thicker wall would slow down this exchange. The huge number and total surface area of capillaries further increase the efficiency of exchange in the tissues.

Question: 4. What are LUB and DUB?

Answer: LUB and DUB are the two sounds of a heartbeat heard with a stethoscope. LUB is the first, louder and longer sound, produced when the atrioventricular (tricuspid and bicuspid) valves close at the start of ventricular contraction. DUB is the second, shorter and sharper sound, produced when the semilunar valves close at the end of ventricular contraction. These sounds indicate the proper working of the heart valves.

Question: 5. Why circulatory system of arthropods is called open type?

Answer: The circulatory system of arthropods is called open because their blood is not always confined to vessels. The heart pumps blood through short arteries into open spaces (sinuses) of the body cavity, where it directly bathes the internal organs. There is no distinction between blood and interstitial fluid, so the general body fluid is called haemolymph. The blood then returns slowly to the heart through small openings called ostia.

Question: 6. Why blood flow faster in arteries?

Answer: Blood flows faster in arteries because they receive blood directly from the pumping heart under high pressure. Arteries have narrower lumen relative to veins and thick elastic muscular walls that maintain this pressure. The total cross-sectional area of arteries is much less than that of capillaries, so velocity stays high. As blood reaches the wide capillary network and then veins, its speed and pressure decrease considerably.

Question: 7. Distinguish between single circuit circulation and double circuit circulation.

Answer: In single circuit circulation, as in fishes, blood passes through the heart only once in one complete circuit of the body, and the heart is two-chambered. In double circuit circulation, as in amphibians to mammals, blood passes through the heart twice: once in the pulmonary circuit to the lungs and once in the systemic circuit to the body. Double circulation keeps oxygenated and deoxygenated blood largely separate in birds and mammals. It supplies oxygen to tissues more efficiently than single circulation.

3. Write detailed answers of the following questions:

Question: 1. Describe the structure of blood vessels in man.

Answer: Human blood vessels are of three types: arteries, veins and capillaries. Arteries have thick, muscular and elastic walls with a narrow lumen to carry blood away from the heart under high pressure. Veins have thinner walls, a wider lumen and valves to prevent backflow, carrying blood towards the heart under low pressure. Capillaries, connecting arterioles and venules, have a wall of only one layer of endothelium for exchange of materials. Each vessel type is structurally suited to its function in circulation.

Question: 2. Describe the structure of human heart.

Answer: The human heart is a somewhat conical, muscular organ enclosed in the pericardium. It consists of four chambers: two upper thin-walled atria and two lower thick-walled ventricles. The right side is separated from the left by inter-atrial and inter-ventricular septa, so oxygenated and deoxygenated blood do not mix. Tricuspid and bicuspid valves guard the atrioventricular openings, and semilunar valves guard the aorta and pulmonary artery. The heart wall is made of cardiac muscle (myocardium) covered externally and lined internally, and the SA node in the right atrium acts as pacemaker.

Question: 3. What is circulation? Explain needs of circulation

Answer: Circulation is the transport of materials within the body through the heart and blood vessels, and in many animals through a lymphatic system as well. It is needed because cells continuously require oxygen and nutrients for metabolism and produce wastes that must be removed. Circulation also distributes hormones, maintains body temperature, keeps pH and fluid balance, and carries antibodies and white blood cells for defence. In large animals diffusion alone is too slow, so circulation is essential for survival of every cell.

Question: 4. State the phases of heartbeat in man.

Answer: The heartbeat consists of a cardiac cycle with three main phases. In atrial systole, both atria contract and push blood into the ventricles. In ventricular systole, the ventricles contract, atrioventricular valves close producing the LUB sound, and blood is pumped into the aorta and pulmonary artery; semilunar valves then close producing the DUB sound. In general diastole, the whole heart relaxes and fills with blood from the veins. The cycle repeats about 72 times per minute in a resting man.

Question: 5. Describe the flow of blood through human heart as regulated by the valves.

Answer: Deoxygenated blood from the body enters the right atrium through the vena cava, and oxygenated blood from lungs enters the left atrium through pulmonary veins. As the atria contract, blood passes through the open tricuspid valve into the right ventricle and through the bicuspid (mitral) valve into the left ventricle. When ventricles contract, these valves close to prevent backflow, and blood leaves through the semilunar valves into the pulmonary artery and aorta respectively. Semilunar valves then close so blood cannot return to the ventricles. Thus valves ensure one-way flow through the heart.

Question: 6. Describe the lymphatic system of man

Answer: The lymphatic system of man consists of lymph capillaries, lymph vessels, lymph nodes and lymphoid tissues such as the spleen. Lymph capillaries collect excess interstitial fluid and absorbed fats from tissues (in villi they are called lacteals) and form lymph. Lymph flows through larger vessels, which have one-way valves and are squeezed by nearby muscle contractions, and passes through lymph nodes where it is filtered and lymphocytes are produced. Finally lymph returns to the blood through the thoracic duct and the right lymphatic duct. This system defends the body and maintains fluid balance. (Diagram to be drawn as in the textbook if required.)

Question: 7. Explain the electrocardiogram with the help of diagram.

Answer: An electrocardiogram (ECG) is the graphical record of the electrical activity of the heart during a cardiac cycle, measured by electrodes placed on the body. A normal ECG wave shows the P wave, produced by atrial depolarization and contraction, followed by the QRS

complex, produced by ventricular depolarization. The T wave represents ventricular repolarization and relaxation. The shape and intervals of these waves show the rate, rhythm and conduction of impulses from the SA node. Any abnormality in the ECG helps diagnose heart disorders.

(Diagram of a normal ECG wave to be drawn as in the textbook.)

Question: 8. Describe pulmonary circulation and systemic circulation

Answer: Pulmonary circulation carries deoxygenated blood from the right ventricle through the pulmonary artery to the lungs, where it is oxygenated, and returns it through the pulmonary veins to the left atrium. Systemic circulation carries oxygenated blood from the left ventricle through the aorta to all body tissues and returns deoxygenated blood through the vena cava to the right atrium. Together they form the double circulation of man. Pulmonary circulation serves gaseous exchange, while systemic circulation supplies the whole body.

Chapter 13: Immunity

1. Encircle the correct choice

(i) The study of functioning and disorders of immune system is termed as:

(a) Immunity

(b) Immune response

✓ (c) Immunology

(d) all of these

Reason: The study of the functioning and disorders of the immune system is termed immunology.

(ii) It is responsible for specific immune response:

(a) Innate Immune system

(b) First line of defense

(c) Second line of defense

✓ (d) Third line of defense

Reason: The third line of defense, consisting of T and B lymphocytes and antibodies, is responsible for the specific immune response.

(iii) Following serves as physical barrier:

(a) Macrophage

✓ (b) Mucous membrane

(c) HCl

(d) Saliva

Reason: Mucous membrane serves as a physical barrier (along with skin) in the first line of defense.

(iv) Our digestive tract consists of what kind of barriers?

(a) Physical barriers

(b) Biochemical barriers

✓ (c) Both a & b

(d) None of these

Reason: The digestive tract has both physical barriers (mucous lining) and biochemical barriers (HCl, enzymes, saliva), so both a & b.

(v) Which of the following is involved in phagocytosis of bacteria?

(a) Neutrophil

(b) Macrophage

✓ (c) Both a & b

(d) None of these

Reason: Both neutrophils and macrophages are phagocytes that engulf and destroy bacteria.

(vi) It is not involved in antigen presentation:

(a) Macrophage

(b) Dendritic cell

✓ (c) Neutrophil

(d) Both a & b

Reason: Neutrophils are phagocytic but are not involved in antigen presentation; macrophages and dendritic cells present antigens.

(vii) The type of receptor exhibited by all nucleated cells other than macrophages is:

✓ (a) MHC-I

(b) MHC-II

(c) Both a & b

(d) TCR

Reason: All nucleated cells exhibit MHC-I receptors, while MHC-II is found on specialized antigen-presenting cells like macrophages.

(viii) Any foreign substance that can elicit immune response is called:

(a) Antibody

✓ (b) Antigen

(c) Both a & c

(d) Antihistamine

Reason: Any foreign substance that can elicit an immune response is called an antigen.

(ix) The type of immunity through the transfer of antibodies from mother to her fetus is:

(a) Natural Active Immunity

✓ (b) Natural Passive Immunity

(c) Artificial Active Immunity

(d) Artificial Passive Immunity

Reason: Transfer of ready-made antibodies from mother to fetus is natural passive immunity, as the fetus's own immune system does not produce them.

(x) Antibodies are secreted by:

(a) T cells

(b) B cells

✓ (c) Plasma cells

(d) Macrophages

Reason: Antibodies are secreted by plasma cells, which are activated B cells transformed into antibody-producing cells.

2. Write short answers of the following questions:

Question: 1. List out any six biochemical barriers.

Answer: Six biochemical barriers of the body are: hydrochloric acid (HCl) in the stomach that kills germs, lysozyme enzyme in tears and saliva that dissolves bacterial walls, and acidic secretions of the skin and sweat that check microbial growth. Bile and digestive enzymes destroy many pathogens entering with food, mucus traps microbes, and interferons and other antimicrobial proteins inhibit viruses and bacteria. These chemical defences form part of the first line of defense. They act non-specifically against any pathogen trying to enter the body.

Question: 2. How the tumor cells are dealt with by our immune system?

Answer: The immune system constantly surveys the body for abnormal cells, including tumor cells, which display altered antigens on their surface. Natural killer (NK) cells and cytotoxic T cells (Tc) recognize these abnormal cells as non-self. They kill them by cytotoxicity, using perforin proteins that make pores in the target cell membrane and granzymes that induce its death. Macrophages also engulf dead tumor cells. This immune surveillance prevents many tumors from developing.

Question: 3. Differentiate between NK Cell and Tc cell.

Answer: NK (natural killer) cells are part of the innate, non-specific second line of defense; they kill virus-infected and tumor cells without prior sensitization and without antigen presentation. Tc (cytotoxic T) cells belong to the specific third line of defense; they are activated by specific antigens presented with MHC-I and then kill the target cells. Both kill by perforin and granzymes. The main difference is specificity and the need for activation: Tc cells are antigen-specific and have memory, NK cells are not.

Question: 4. List out the ways of second line of defense.

Answer: The second line of defense acts when pathogens cross the first line. Its ways include phagocytosis by neutrophils, monocytes/macrophages and dendritic cells, and killing by natural killer cells using perforin and granzymes. Inflammation brings more blood and phagocytes to the infected area, and fever inhibits microbial growth and speeds up defence reactions. Antimicrobial proteins and the complement system also attack invading microbes. These defences are non-specific but act rapidly.

Question: 5. Even though the core proteins are the same, how antibodies differ from each other?

Answer: All antibodies have the same basic Y-shaped structure and constant core regions, but they differ in their variable regions. The variable regions at the tips of the arms have different amino acid sequences, forming antigen-binding sites of different shapes. Because of this, each antibody binds a specific antigen. Antibodies also differ in their class (type of heavy chain), which determines their function and location, e.g. in blood, secretions or on mast cells.

Question: 6. What is antipyretic therapy and why it is used?

Answer: Antipyretic therapy is the use of medicines that lower body temperature during fever, e.g. paracetamol. It is used when fever becomes very high or prolonged, because excessive fever can damage body proteins, cause discomfort, dehydration and convulsions, especially in children. By reducing the raised set-point of temperature in the hypothalamus, antipyretics bring relief. Mild fever itself is protective, so antipyretics are used mainly to control harmful effects of high fever.

Question: 7. List out four autoimmune disorders of man.

Answer: Four autoimmune disorders of man are: type-I diabetes mellitus, in which immune cells destroy insulin-producing cells of the pancreas; rheumatoid arthritis, in which joints are attacked causing pain and deformity; psoriasis, a skin disorder with rapid abnormal skin growth; and multiple sclerosis / myasthenia-type disorders in which nervous or muscle tissues are attacked. In all these, the immune system fails to recognize its own tissues as self. Misdirected self-antibodies destroy the body's own cells.

Question: 8. What is phagocytosis? Name some WBCs acting as phagocytes.

Answer: Phagocytosis is the process by which certain white blood cells engulf and digest pathogens and dead material. The phagocyte extends pseudopodia around the microbe, encloses it in a vacuole, and lysosomal enzymes kill and digest it. The main WBCs acting as phagocytes are neutrophils, monocytes (which become macrophages in tissues) and dendritic cells. Eosinophils also show limited phagocytic action. Phagocytosis is a major part of the second line of defense.

Question: 9. What is inflammation?

Answer: Inflammation is the local protective response of tissues to infection or injury. Damaged cells and mast cells release histamine and other chemicals, which dilate blood vessels and increase their permeability. The area becomes red, warm, swollen and painful, and more blood with phagocytes reaches the site. Phagocytes destroy the pathogens and clean the debris, helping healing. Thus inflammation limits infection and starts tissue repair.

Question: 10. Outline the harmful effects of fever.

Answer: Although mild fever is protective, high and prolonged fever is harmful. It increases metabolic rate and water loss, causing dehydration, weakness and discomfort. Very high temperature can denature enzymes and damage tissues, especially of the brain. In children, sudden high fever may cause convulsions (fits). Prolonged fever also reduces appetite and can disturb heart and kidney function, so it must be controlled.

3. Write detailed answers of the following questions:

Question: 1. State and explain how a bacterium is identified as non-self by our immune system.

Answer: A bacterium is identified as non-self because its surface carries antigens and molecular patterns that are different from the body's own markers. Antigen-presenting cells such as macrophages and dendritic cells engulf the bacterium and display its antigen fragments on their surface with MHC-II molecules. Helper T cells recognize this antigen-MHC combination as foreign, because body cells display only self antigens with MHC. This recognition activates the specific immune response. Self cells are protected because lymphocytes that react against self antigens are normally eliminated or inactivated.

Question: 2. What is cytotoxicity? Compare and contrast the mechanisms of NK cell and Tc cell.

Answer: Cytotoxicity is the killing of target cells (infected or tumor cells) by immune cells through toxic substances. Both NK cells and Tc cells kill by releasing perforin, which makes pores in the target cell membrane, and granzymes, which enter through these pores and induce the target cell to die. The difference is that NK cells act non-specifically and immediately, without antigen presentation or prior activation, as part of innate

immunity. Tc cells act specifically after recognizing a particular antigen presented with MHC-I, are part of adaptive immunity, and form memory cells.

Question: 3. What is inflammation? Why it is developed and how it is controlled?

Answer: Inflammation is the body's local response to infection or tissue injury, marked by redness, heat, swelling and pain. It develops because damaged cells, mast cells and basophils release histamine and other mediators that dilate vessels and make them leaky, so plasma and phagocytes accumulate at the site to destroy pathogens and repair tissue. It is controlled by anti-inflammatory mechanisms: once the pathogen is removed, mediator release stops, antihistamines and steroids can reduce excessive response, and macrophages clean the area. Controlled inflammation is protective, but excessive inflammation can damage tissues.

Question: 4. Explain the underlying mechanism of the fever. State and explain its protective role as well as harmful effects.

Answer: Fever develops when pyrogens released by pathogens and immune cells act on the temperature centre (hypothalamus) and raise its set-point. The body then produces and conserves more heat by shivering and vasoconstriction until the new temperature is reached. Protectively, raised temperature inhibits the growth of many bacteria and viruses and speeds up phagocytosis and immune reactions. Harmfully, very high or prolonged fever causes dehydration, enzyme damage, discomfort and convulsions in children, so it needs monitoring and, when excessive, antipyretic treatment.

Question: 5. Discuss the structural features of antibody? How it is helpful in getting rid of pathogenic organisms.

Answer: An antibody is a Y-shaped protein made of four polypeptide chains: two identical heavy chains and two identical light chains, joined by disulfide bonds. The stem and lower parts form constant regions, while the tips of the arms have variable regions that form two identical antigen-binding sites, specific for one antigen. Antibodies get rid of pathogens by binding their antigens, neutralizing toxins and viruses, clumping (agglutinating) microbes, and marking them for phagocytosis (opsonization). They also activate complement, which lyses the pathogen. Plasma cells secrete these antibodies during the immune response.

Question: 6. What do you understand by autoimmune disorder? Explain the role of T cells and B cells in transplant rejection.

Answer: An autoimmune disorder is an abnormality in which the immune system fails to recognize one or more of its own tissues as self and attacks them, producing self-antibodies that destroy own cells, e.g. type-I diabetes and rheumatoid arthritis. In transplant rejection, the recipient's helper T cells recognize the donor organ as foreign through antigen-presenting cells, proliferate, and activate NK cells and cytotoxic T cells that directly attack the grafted tissue (cell-mediated rejection). The activated helper T cells also signal B cells to become plasma cells whose antibodies destroy the graft (humoral rejection). For this reason transplant patients are given immunosuppressant drugs.

Chapter 14: Gaseous Exchange

1. Encircle the correct choice

(i) In which part of the respiratory system, gaseous exchange takes place?

✓ (a) Alveoli

(b) Pharynx

(c) Larynx

(d) Trachea

Reason: Gaseous exchange takes place in the alveoli, whose thin moist walls are surrounded by blood capillaries.

(ii) Which of the following statements is true about involuntary breathing?

(a) It is controlled by the bronchioles

(b) It is controlled by the pulmonary arterioles

(c) It is controlled by the alveolar-capillary network

✓ (d) It is controlled by the neurons, located in the medulla and pons

Reason: Involuntary breathing is controlled by neurons located in the respiratory centres of the medulla and pons.

(iii) The tiny air sacs present in human lungs is called.

✓ (a) Alveoli

(b) Bronchus

(c) Bronchioles

(d) All of the above

Reason: The tiny air sacs present in human lungs are called alveoli.

(iv) The exchange of gases between the external environment and the lungs.

(a) Respiration

✓ (b) External respiration

(c) Cellular respiration

(d) None of the above

Reason: The exchange of gases between the external environment and the lungs (blood at alveoli) is called external respiration.

(v) The maximum volume of air contained in the lung by a full forced inhalation is called.

(a) Tidal volume

(b) Vital capacity

(c) Ventilation rate

✓ (d) Total lung capacity

Reason: The maximum volume of air contained in the lungs after a full forced inhalation is the total lung capacity.

(vi) Which one of the following is correct regarding larynx?

(a) It houses the vocal cords

(b) It prevents the invading pathogens into the trachea

(c) It is an organ made of cartilage and connects the pharynx to the trachea

✓ (d) All of the above.

Reason: All of the above are correct: the larynx is a cartilage organ connecting pharynx to trachea, houses the vocal cords, and its epiglottis guards the trachea.

(vii) Which of the following is the function of the trachea?

(a) Gaseous Exchange

✓ (b) Filters the air we breathe

(c) Exhales the air from the body

(d) All of the above

Reason: The trachea conducts air and filters the air we breathe through its mucus and cilia, which trap dust and germs.

(viii) Which of the following organs functions as an air conditioner?

(a) Larynx

(b) Pharynx



(c) Nasal chambers

(d) All of the above

Reason: The nasal chambers function as an air conditioner, warming, moistening and filtering the incoming air.

(ix) Which of the following statements is true about the entry of air into the lungs?



(a) Air enters the body and travels to the lungs through the mouth and the nose

(b) Air enters the body and travels to the lungs through the esophagus and gullet

(c) Air enters the body and travels to the lungs through the windpipe and the pores

(d) Air enters the body and travels to the lungs through the nose and the nervous system.

Reason: Air enters the body through the mouth and the nose and travels via the windpipe (trachea) to the lungs.

2. Write short answers of the following questions:

Question: 1. When breathing process develop in animals?

Answer: The breathing process developed in animals when they became

large and multicellular and simple diffusion through the body surface was no longer sufficient for their needs. Aquatic animals first developed gills for extracting oxygen from water. With the invasion of land, lungs and tracheal systems evolved to take oxygen directly from air. Thus breathing mechanisms developed step by step with increasing body size, activity and terrestrial life.

Question: 2. What type of ventilation occur in human?

Answer: Humans show pulmonary ventilation by negative pressure breathing (also described as tidal ventilation). Air is drawn into the lungs by expanding the chest cavity with the diaphragm and intercostal muscles, and the same pathway is used for air leaving the lungs. The same air passage therefore works in both directions, and fresh air mixes with some residual air in the lungs. Ventilation rate in a healthy adult is about 15-18 breaths per minute at rest.

Question: 3. Why rate of breathing increases in human beings?

Answer: The rate of breathing increases when the body needs more oxygen and produces more carbon dioxide, e.g. during exercise, exertion, fever or stress. Increased CO2 and H+ concentration in the blood is detected by chemoreceptors in the aortic and carotid bodies. They send signals to the respiratory centre in the medulla, which increases the rate and depth of breathing. This removes extra CO2 and restores normal blood pH and oxygen supply.

Question: 4. Why breathing of human being called negative pressure breathing?

Answer: Human breathing is called negative pressure breathing because air is sucked into the lungs by creating a pressure inside the lungs lower

than atmospheric pressure. When the diaphragm contracts and flattens and the rib cage expands, the volume of the chest cavity increases and pressure in the lungs falls below atmospheric pressure. Air therefore rushes in from outside to equalize the pressure. During expiration the process reverses and air is pushed out.

Question: 5. How CO2 from cell transport to lungs?

Answer: Carbon dioxide produced in cells diffuses into blood capillaries and is transported to the lungs in three ways: a small part dissolves in plasma, a part binds with haemoglobin as carbaminohaemoglobin, and the major part is carried as bicarbonate ions formed with the help of carbonic anhydrase in red blood cells. Blood carries it through veins to the heart and then through the pulmonary artery to the lungs. At the alveoli, CO2 is released and exhaled.

Question: 6. Which of the pigment is called respiratory pigment?

Answer: Haemoglobin is called the respiratory pigment in humans and many animals. It is a red, iron-containing protein present in red blood cells. Each haemoglobin molecule can carry up to four oxygen molecules, forming oxyhaemoglobin, and transports oxygen from the lungs to the tissues. About 97% of oxygen in the body is transported by haemoglobin, while only about 3% is carried dissolved in plasma.

Question: 7. Draw a flow chart for the passage of air from external nares to alveoli.

Answer: The passage of air is: External nares → nasal cavity → internal nares → pharynx → larynx → trachea → primary bronchi → secondary bronchi → tertiary bronchi → bronchioles → alveolar ducts → alveoli. Air is filtered, warmed and moistened in the nasal cavity and trachea on its way.

At the alveoli, gaseous exchange with blood capillaries takes place. (To be written/drawn as a flow chart in the notebook.)

Question: 8. Why hair and mucus glands are present in nostrils and trachea?

Answer: Hair in the nostrils traps large dust particles and prevents them from entering the air passages. Mucus glands in the nostrils and trachea secrete sticky mucus that traps fine dust, pollen and germs carried by inhaled air. Cilia on the lining then sweep the mucus with trapped particles towards the pharynx, from where it is swallowed or expelled. Together, hair and mucus keep the respiratory surface clean and protect the lungs from infection and damage.

Question: 9. How exchange of gases occurs at the alveolar level?

Answer: At the alveolar level, exchange occurs by simple diffusion across the very thin (about 1 micrometre) barrier between alveolar air and blood capillaries. Oxygen concentration is higher in alveoli than in the incoming blood, so oxygen diffuses into the blood and binds with haemoglobin. Carbon dioxide concentration is higher in blood than in alveoli, so it diffuses out into the alveoli and is exhaled. The large surface area, moist surface and rich capillary network make this exchange very rapid and efficient.

3. Write detailed answers of the following questions:

Question: 1. What is respiration? Describe human respiratory system?

Answer: Respiration is the process of exchanging gases with the environment and releasing energy from food; breathing is its mechanical part, consisting of inhalation and exhalation. The human respiratory system comprises the nasal cavity, pharynx, larynx, trachea, bronchi,

bronchioles and lungs containing millions of alveoli. Air passes through this passage to the alveoli, where oxygen diffuses into blood and CO2 diffuses out. The diaphragm, ribs and intercostal muscles bring about breathing movements, and the lungs are protected by pleural membranes.

Question: 2. What are the 5 main functions of the respiratory system?
Answer: The five main functions of the respiratory system are: (1) gaseous exchange — taking in oxygen and removing carbon dioxide; (2) supplying oxygen for cellular respiration and energy production in the body; (3) helping in regulation of blood pH by controlling CO2 level; (4) production of sound through the vocal cords in the larynx; and (5) protection of air passages by filtering, warming and moistening air and trapping germs in mucus, as well as a sense of smell in the nasal cavity. These functions keep the internal environment stable.

Question: 3. Draw labelled diagram of human respiratory system.
Answer: The human respiratory system is drawn showing the nasal cavity at the top, followed by the pharynx, larynx and the long trachea supported by C-shaped cartilage rings. The trachea divides into two primary bronchi, one entering each lung, which branch into smaller bronchi and bronchioles ending in alveoli. The right lung has three lobes and the left lung two lobes, and below the lungs lies the dome-shaped diaphragm. (Labelled diagram to be drawn as in the textbook: nasal cavity, pharynx, larynx, trachea, bronchi, lungs, alveoli and diaphragm labelled.)

Question: 4. Describe the three regions of the pharynx and their functions.
Answer: The pharynx is a common passage for air and food divided into three regions. The nasopharynx lies behind the nasal cavity and only passes air, containing openings of the auditory (Eustachian) tubes that

equalize air pressure. The oropharynx lies behind the oral cavity and passes both air and food, and contains the tonsils that help in defence. The laryngopharynx is the lowest part, leading to the larynx for air and the oesophagus for food, and the epiglottis directs food away from the windpipe during swallowing.

Question: 5. What is the mechanism of inspiration and expiration?

Answer: Inspiration is an active process: the diaphragm contracts and flattens and the external intercostal muscles lift the ribs up and out, increasing the volume of the chest cavity. Pressure inside the lungs falls below atmospheric pressure, so air rushes in through the air passage to the alveoli. Expiration is normally passive: the diaphragm and intercostal muscles relax, the chest cavity decreases in volume, and increased pressure pushes air out of the lungs. One complete inspiration and expiration forms one breath cycle.

Question: 6. Describe three ways in which carbon dioxide can be transported.

Answer: Carbon dioxide is transported from tissues to lungs in three ways. First, about 5-7% is carried dissolved in the blood plasma. Second, about 20-23% binds with haemoglobin in red blood cells to form carbaminohaemoglobin. Third, the major part (about 70%) is converted in RBCs, with the enzyme carbonic anhydrase, into carbonic acid which dissociates into bicarbonate ions (HCO_3^-) and H^+; bicarbonate is carried in plasma as $KHCO_3$ and $NaHCO_3$. At the lungs these reactions are reversed and CO_2 is released for exhalation.